FROM EXCEL TO R

SUND DATA LAB

WHO ARE WE?

Center for Health Data Science (**HeaDS**):

- Data Science Research Groups
- HDS Sandbox (Infrastructure Project)
- **SUND Data Lab**

https://heads.ku.dk/

SUND Data Lab:

- **Courses & Workshops, Seminars, etc.**
- **Health Data Science Consultations**
- **Commissioned Research & Supervision**
- **Hub for Data Science (Matchmaking)**

Diana Andrejeva

Henrike Zschach

Thilde Terkelsen

Helene Wegener

Chelsea Lennox

Tugce Karaderi

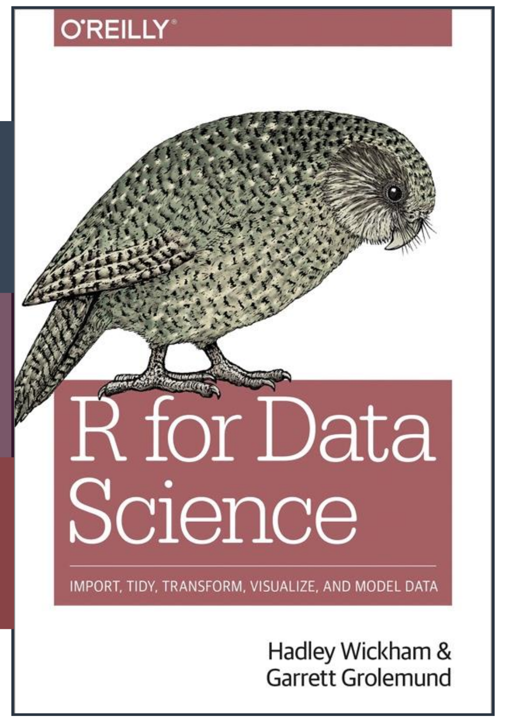
Data Science Laboratory (DSL) - https://datalab.science.ku.dk/

THE PRACTICALS

Two days: 9.00-16.00. There will be coffee breaks, we promise

“R for Data Science” - a generally useful book on R, also for this course

The course is build on hands-on presentations & exercises



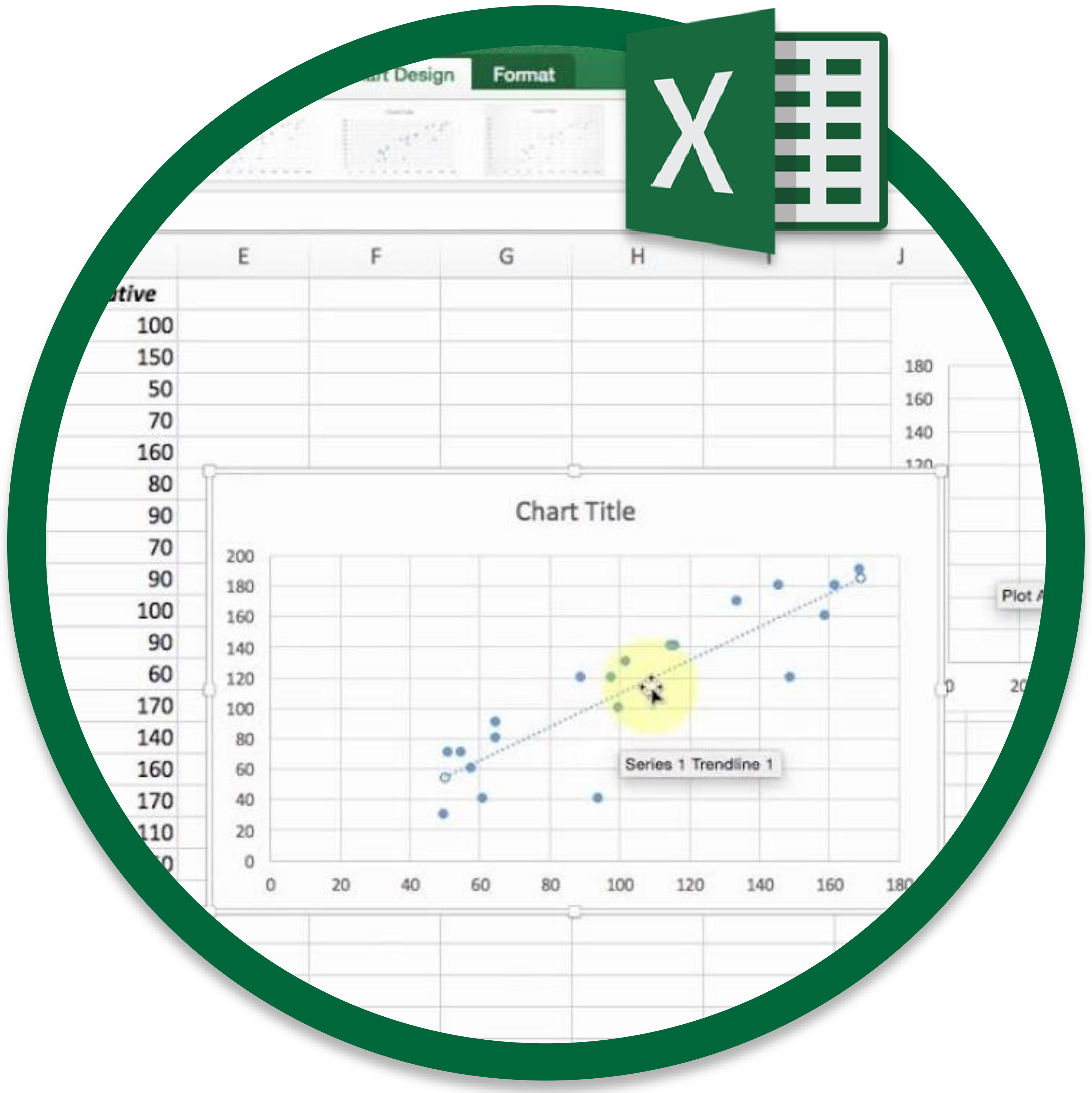
Download and install the newest version of R (https://cran.r-project.org/)

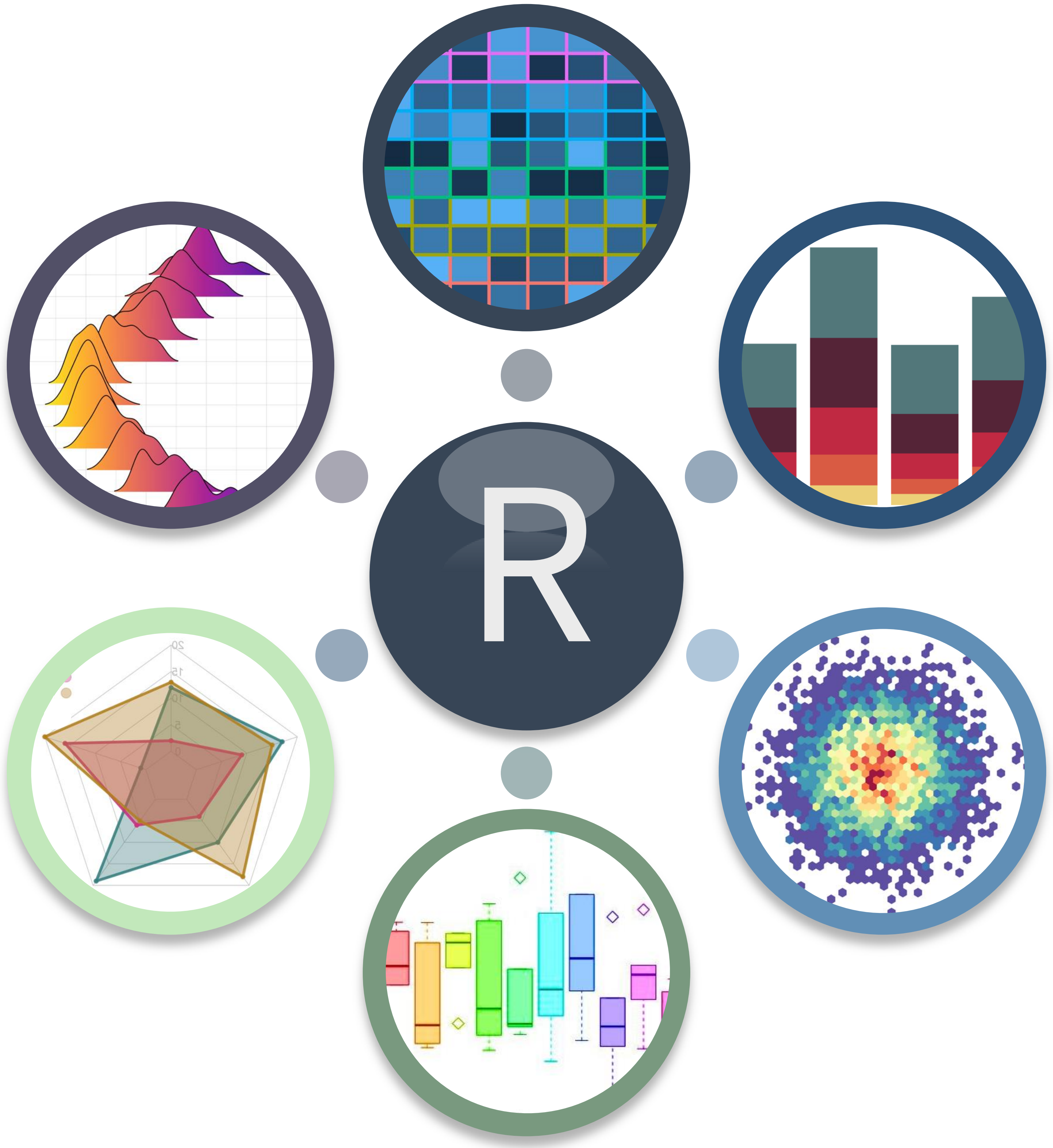
Download and install the newest version of R-studio (https://posit.co/download/rstudio-desktop/)

Go to course website: https://center-for-health-data-science.github.io/FromExceltoR/

WELCOME TO

FROM EXCEL TO R





WHAT WILL I LEARN?

THE BASICS

RStudio
Scripts, paths, files
R project
Help recourses

DATA WRANGLING

tidyverse
Data Structures
Useful Functions
Pipe (“clean” code)

PLOTTING

ggplot2
Code structure
ggplot2 + tidy data

STATISTICS

Linear models
ANOVA
Post Hoc Tests

PROGRAM

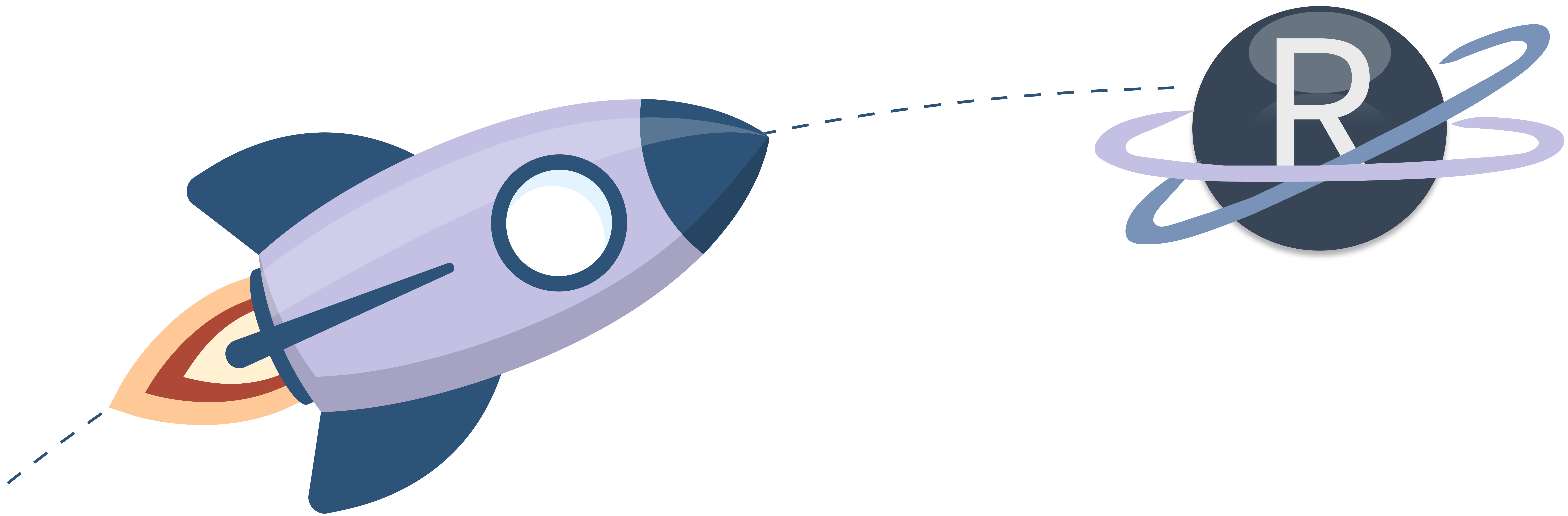
DAY 1	DAY 2
08:30 - Installation Issues + Coffee	08:30 - Coffee + Optional Q&A
09:00 - Introduction to R	09:00 - ggplot2
09:45 - Exercise 0	09:45 - Break
10:15 - Break	10:00 - ggplot2 Exercises
10:30 - Introduction to Quarto	12:00 - Lunch
11:00 - Exercise 1	13:00 - Applied Statistics
12:00 - Lunch	14:30 - Break
13:00 – Tidyverse	14:45 - Data Analysis Exercise
14:00 - Break	15:45 - Cool things in R + Course Evaluation
14:15 - Tidyverse Exercise	16:00 - Bye Bye Bye!
16:00 - See you tomorrow	

PROGRAM

DAY 1	DAY 2
08:30 - Installation Issues + Coffee	08:30 - Coffee + Optional Q&A
09:00 - Introduction to R	09:00 - ggplot2
09:45 - Exercise 0	09:45 - Break
10:15 - Break	10:00 - ggplot2 Exercises
10:30 - Introduction to Quarto	12:00 - Lunch
11:00 - Exercise 1	13:00 - Applied Statistics
12:00 - Lunch	14:30 - Break
13:00 – Tidyverse	14:45 - Data Analysis Exercise
14:00 - Break	15:45 - Cool things in R + Course Evaluation
14:15 - Tidyverse Exercise	16:00 - Bye Bye Bye!
16:00 - See you tomorrow	

FROM EXCEL TO R

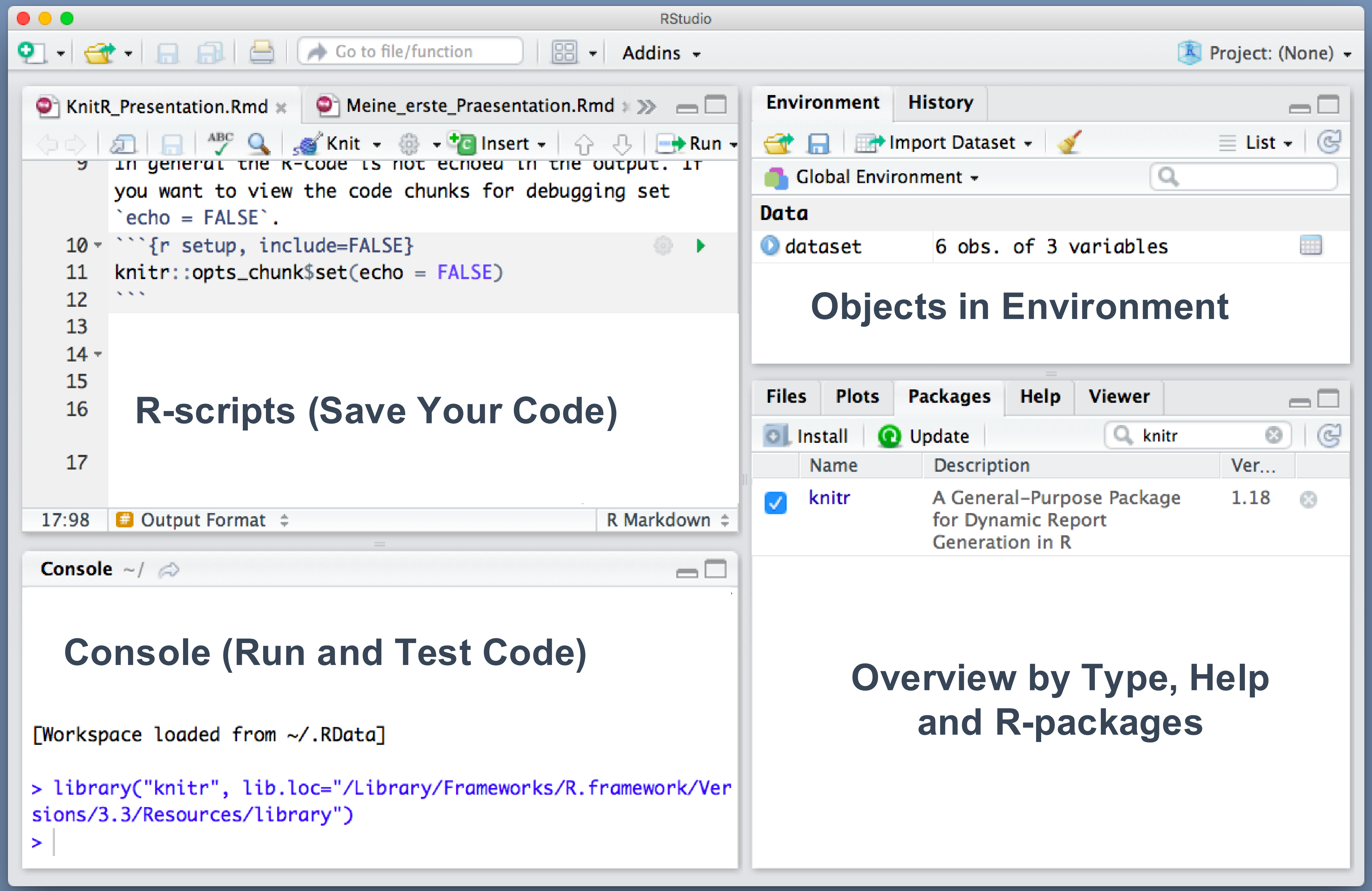
LET'S GET STARTED



R & FRIENDS

Scripting / Programming Language

Reports (html, pdf, word)



R Code Interpreter and Editor

THE ANATOMY OF R

The **programming language** R may be used from within in the **Code Editor** R Studio

The R code you write may be saved in **scripts** with extensions: .R .Rmd .qmd

Functions perform actions based on **input arguments** and return an output:

```
mean(x)
sd(x)
```

Packages are bundles/libraries of functions.

dplyr
```
select()
mutate()
```

ggplot2

readr

purrr

tibble
```
tibble()
```

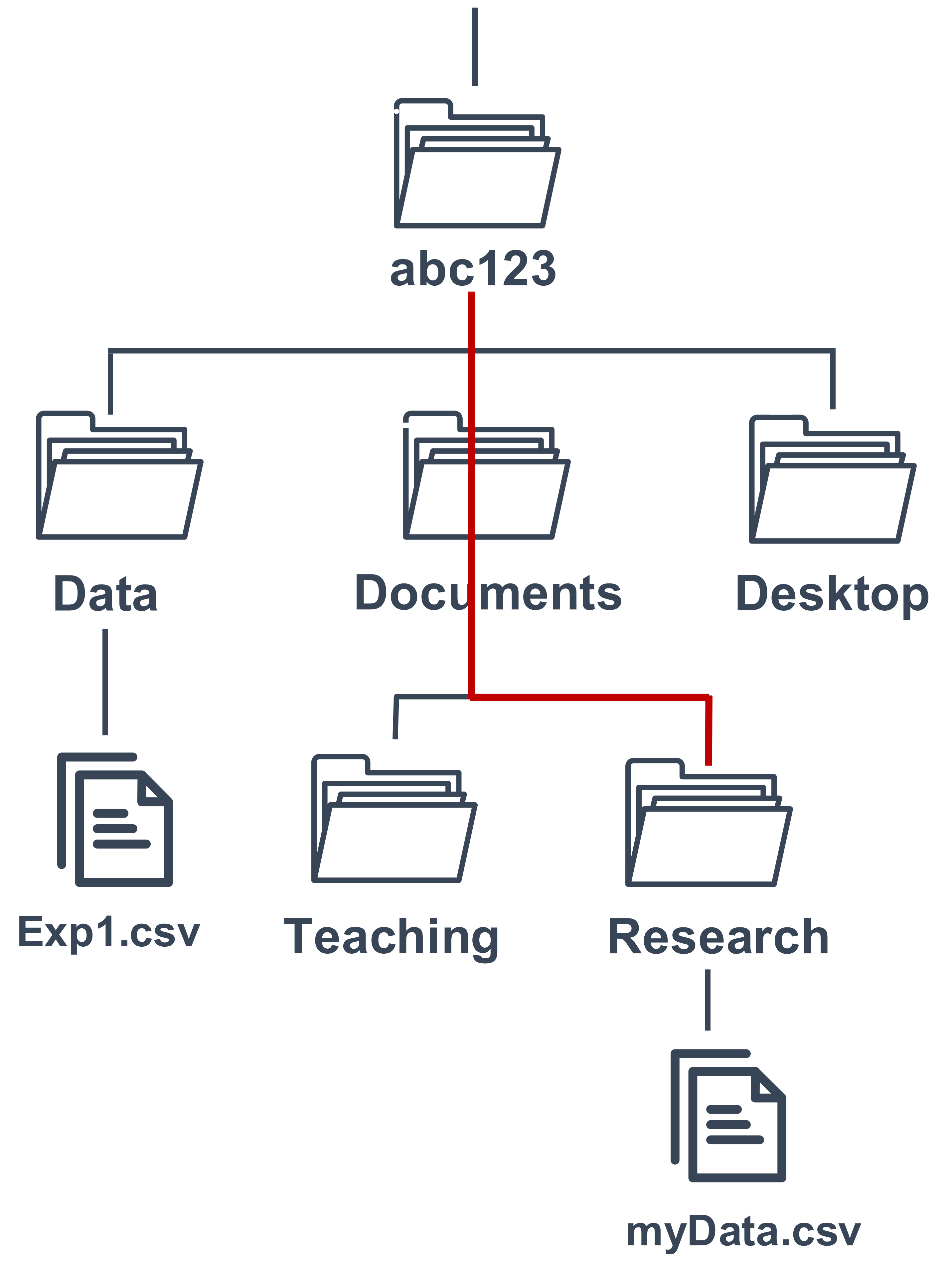
PATHS – LOCATION, LOCATION, LOCATION

- The directories on your computer are organized in a **File Tree**
- **Move** between directories or **point to** a place in the tree with a specific file

We do this by specifying a **path:**

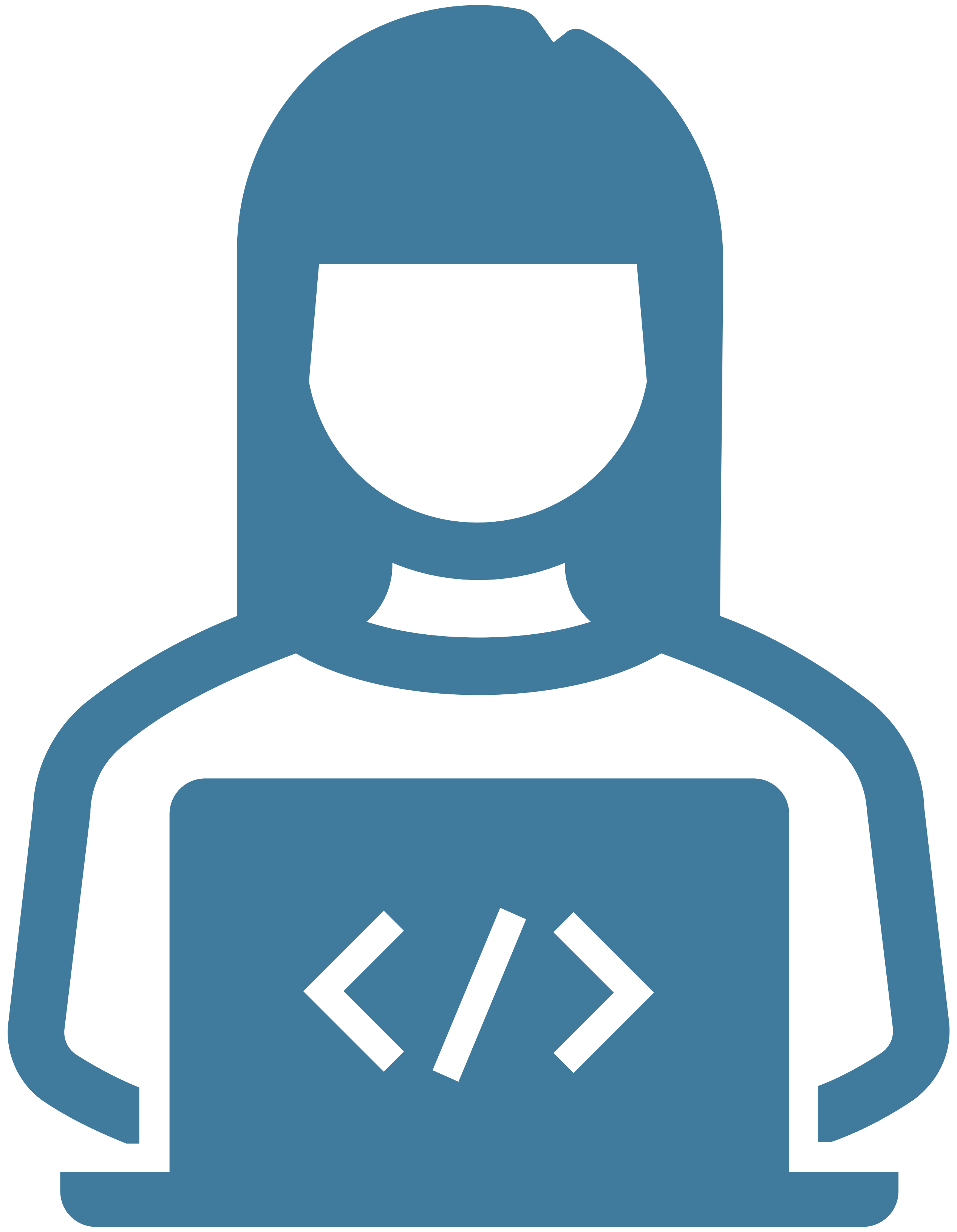
- **Move to:** `/abc123/Documents/Research`
- **Point to:** `/abc123/Documents/Research/myData.csv`

Go through Exercise 0 in plenum!



FROM EXCEL TO R

LIVE CODING 1 – INTRO TO R



QUARTO DOCUMENTS

quarto

- Tool which supports **reproducible** coding/publication (reports)
- Quarto works with a **markup language**
- A markup language is **text-coding system** which specifies structure & format of a document and the text within it
- We **render** (compile) a Quarto → html, pdf, word, website

Pointers for quarto:

- Separate sections with headers
- Do not overfill code chunks
- Write meaningful and precise comments

90% of all code comments:

https://quarto.org

QUARTO CHEAT SHEET

GETTING STARTED

YAML parameter:

```
---
title: My Project Name
output:
      html_document (pdf_document, …)
---
```

Code Chunk:

Source mode:

```
```{r}
# some R code
```
```

Visual mode:

```
{r}
# some R code
```

CHUNK OPTIONS

Code Options:

```
{r echo = FALSE} # don't print code (default is TRUE)
{r eval = FALSE} # don't run code (default is TRUE)
{r error = FALSE} # don't display error message (default is TRUE)
(Can also be set for warning and message)
```

Figure Options:

```
fig.align (='left', 'right', 'center')
fig.cap (= 'my figure cation')
fig.height (= n), fig.width (= n)
```

TEXT

Source mode:
Header

Header size ranging from largest (one #) to smallest (six #):

`# my.text, ## my.text, ### my.text, etc.`

Text

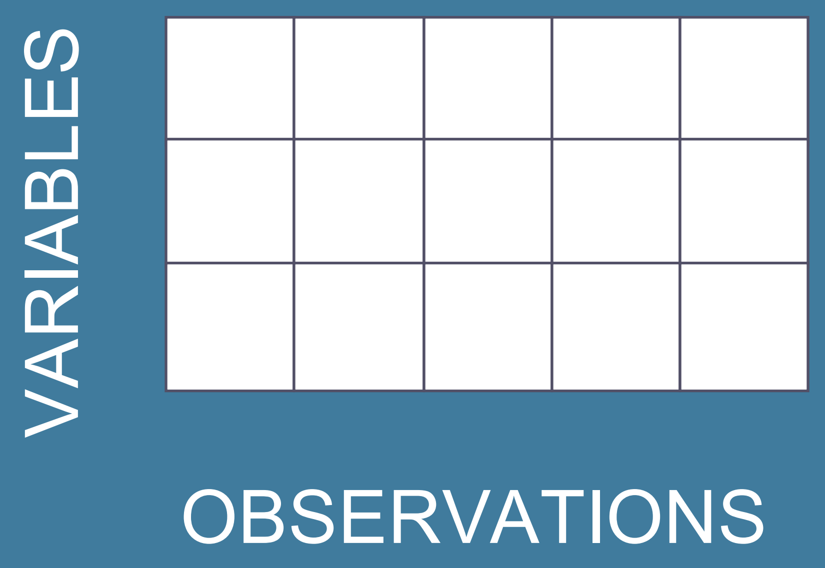
```
*italics*
**bold**
`highlighted`
```

Lists

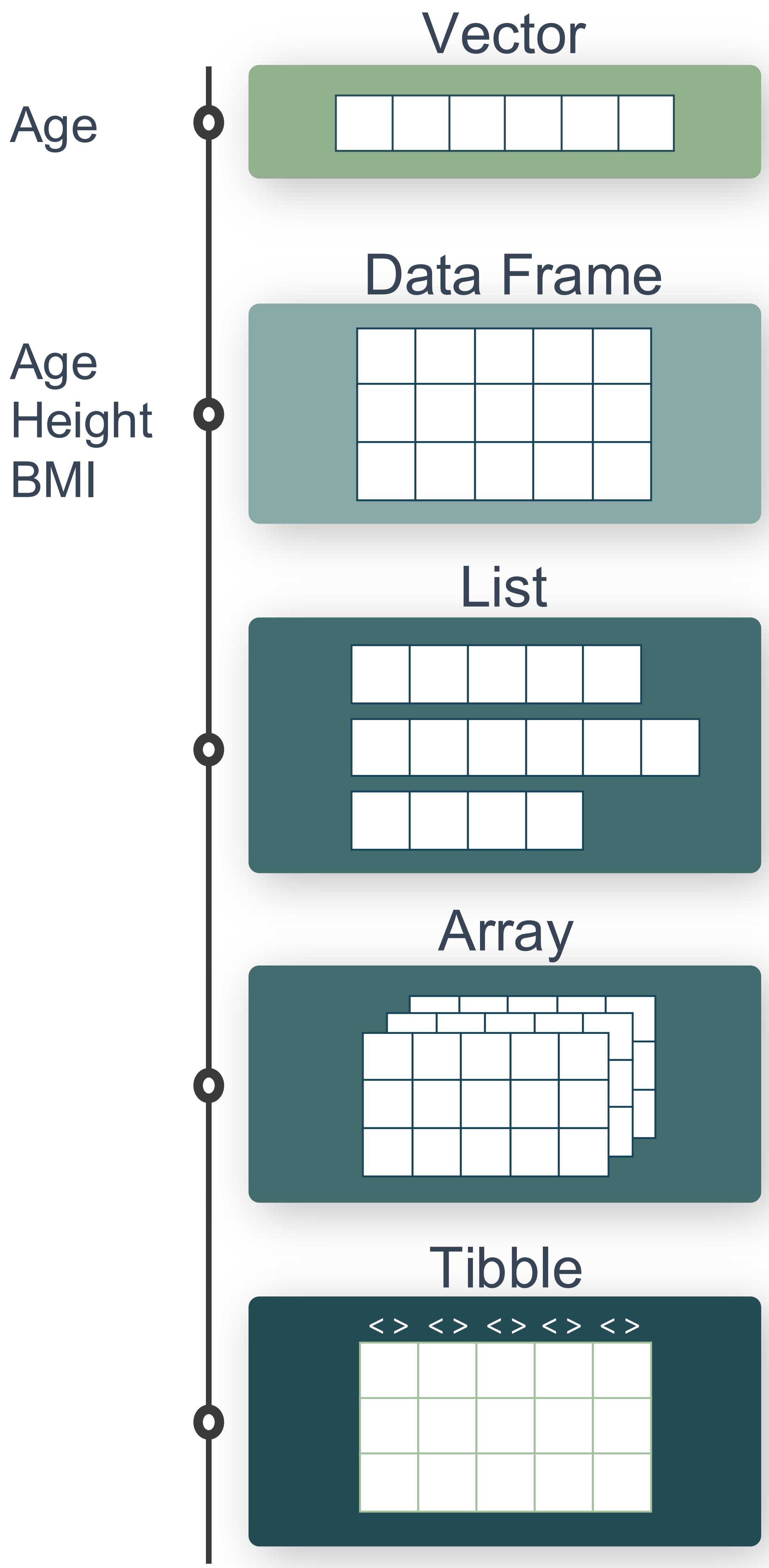
```
* List item1 (filled dot)
    + sub-item1 (open dot)
1. List item1 (numbered)
    i) sub-item1 (roman)
```

FROM EXCEL TO R

R DATA TYPES & STRUCTURES



DATA STRUCTURES



DATA TYPES

Numeric

-1.5, 2.7, 3.2

Integer

-2, -1, 0, 3, 5

Character

“A”, “Bat”, “Tom”

Factor

G1, G1, G2, G3

Logical

TRUE, FALSE

https://www.rstudio.com/wp-content/uploads/2016/10/r-cheat-sheet-3.pdf

INTRO CHEAT SHEET

GETTING STARTED

Basics:

```
getwd() # location
install.packages("package_name") # install packages
library(package_name) # load packages
```

Read in Data:

```
read.xlsx("name.xlsx") # library(readxl)
read.delim("name.txt", sep ="\t")
read.csv("name.csv", sep=";")
```

OVERVIEW

Overview:

```
head(df, n=10), tail(df, n=10) # first or last 10 rows
unique(), table(), count() # unique vals, count vals
view() # view data as table
df$col1 # extract column from dataframe
nrow(df), ncol(df) # number of rows/columns
```

DATA TYPES & CONDITIONS

Type/Class:

```
class() # get data type/class
is.numeric(x), is.character(), is.factor() # is data type
as.numeric(x), as.character(), as.factor() # change data type
```

Conditions:

```
a == b    a != b    a > b    a < b
a >= b    a <= b    is.na()  is.null()
```

STATISTICS & BASE PLOTS

Summary Stats:

```
summary() # summary statistics
mean(), median(), sd(), sum(), log(), max(),
min(), quantile(), var()
```

Plots:

```
plot(x)
plot(x, y) or plot(col1, col2, df) # scatter
hist(x) # histogram
```

ONLINE RESOURCES FOR **R**

https://www.r-project.org/

GET STARTED

https://rseek.org/

https://rstudio.com/resources/cheatsheets/

http://www.cookbook-r.com/

https://www.statmethods.net/r-tutorial/index.html

GRAPHICS

https://www.r-graph-gallery.com/

http://r-statistics.co/Top50-Ggplot2-Visualizations-MasterList-R-Code.html

BOOKS & COURSES

https://www.r-bloggers.com/best-books-to-learn-r-programming/

https://www.datacamp.com/

https://www.codecademy.com/

https://www.coursera.org/

OTHER RESOURCES

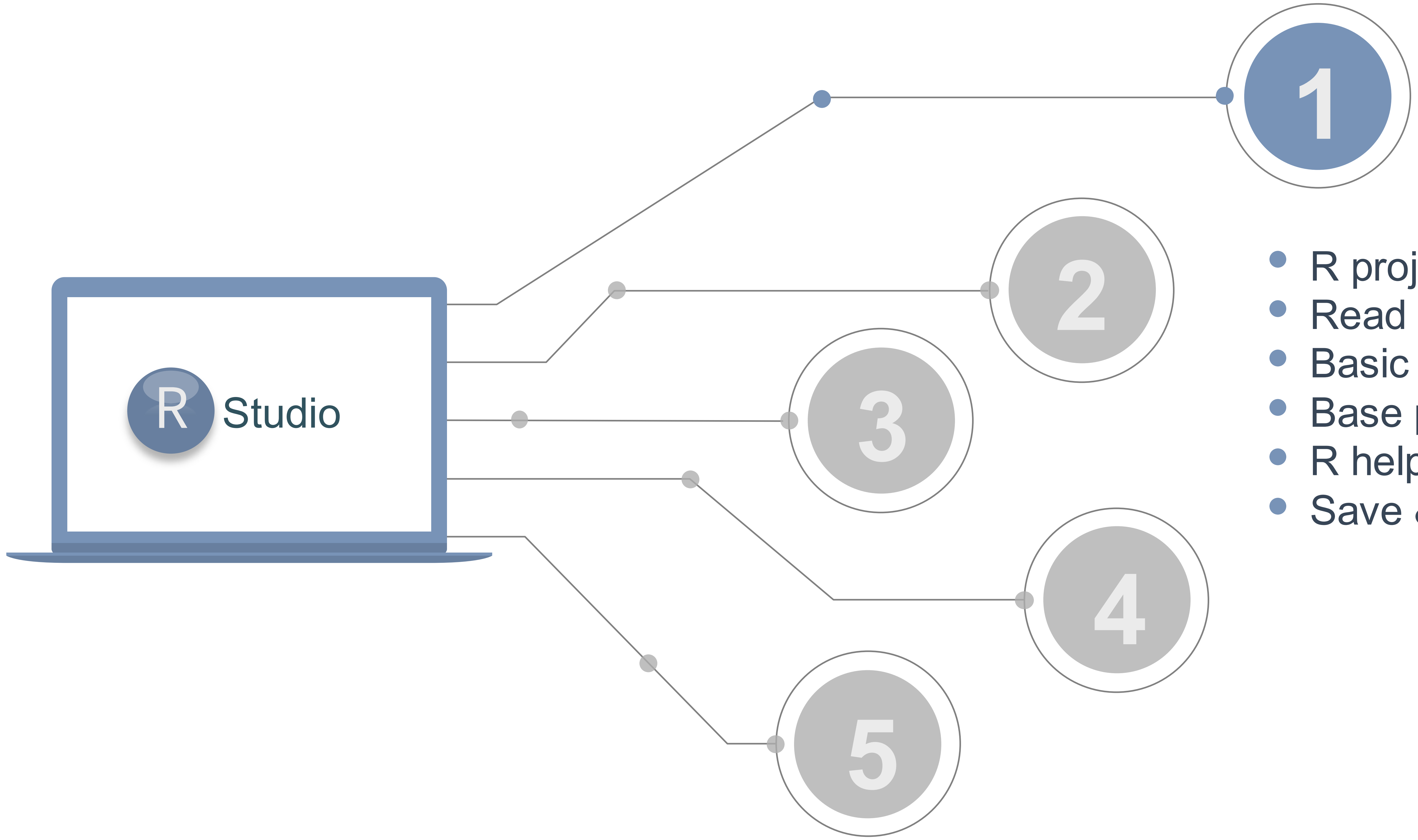
https://github.com/trending/r

https://blog.revolutionanalytics.com/

https://stackoverflow.com/questions/tagged/r

FUNDAMENTALS

EXERCISE 1



- R project & R Script
- Read & Write
- Basic calculus
- Base plot
- R help
- Save & Shutdown

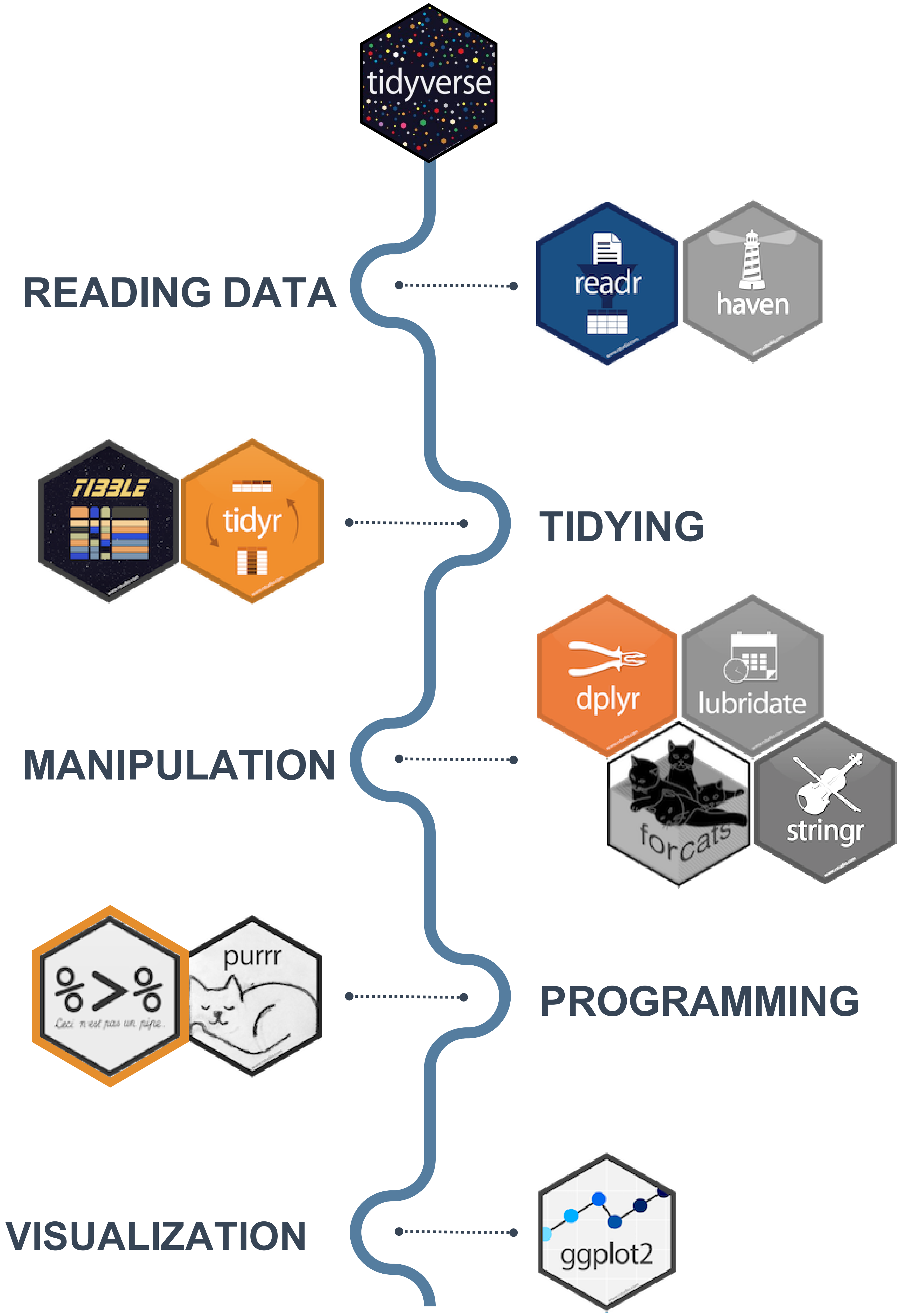
https://www.tidyverse.org/

TIDYVERSE

tidyverse is a collection of R packages for data science

“The packages share an underlying design philosophy, grammar, and data structures.” *Wickham and Grolemund*

tidyverse is used to “tidy up” your datasets, so they are easy to work with

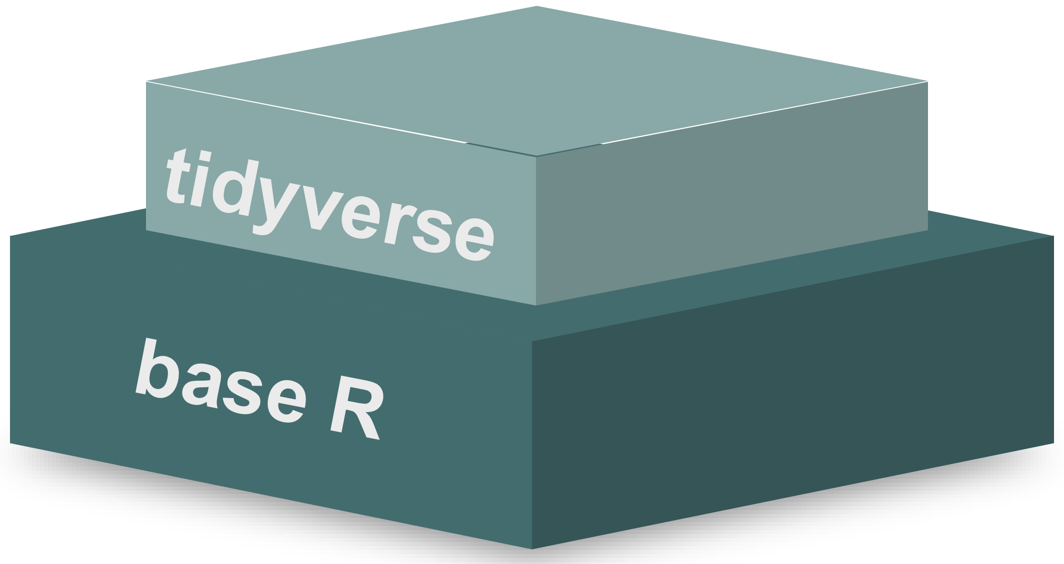


https://www.tidyverse.org/

CECI N'EST PAS UNE PIPE

%>%

- You do NOT have to “chose” between tidyverse and base R



BENEFITS

- Short & well-organised code
- Tidy datasets, easy to work with
- Great documentation
- Functions with logical names & inputs

CONSIDERATIONS

- Can be less stable
- “Different syntax”
- Can be computationally slow

base R

```
# think from the inside out

g(f(x,y),z)
```

tidyverse

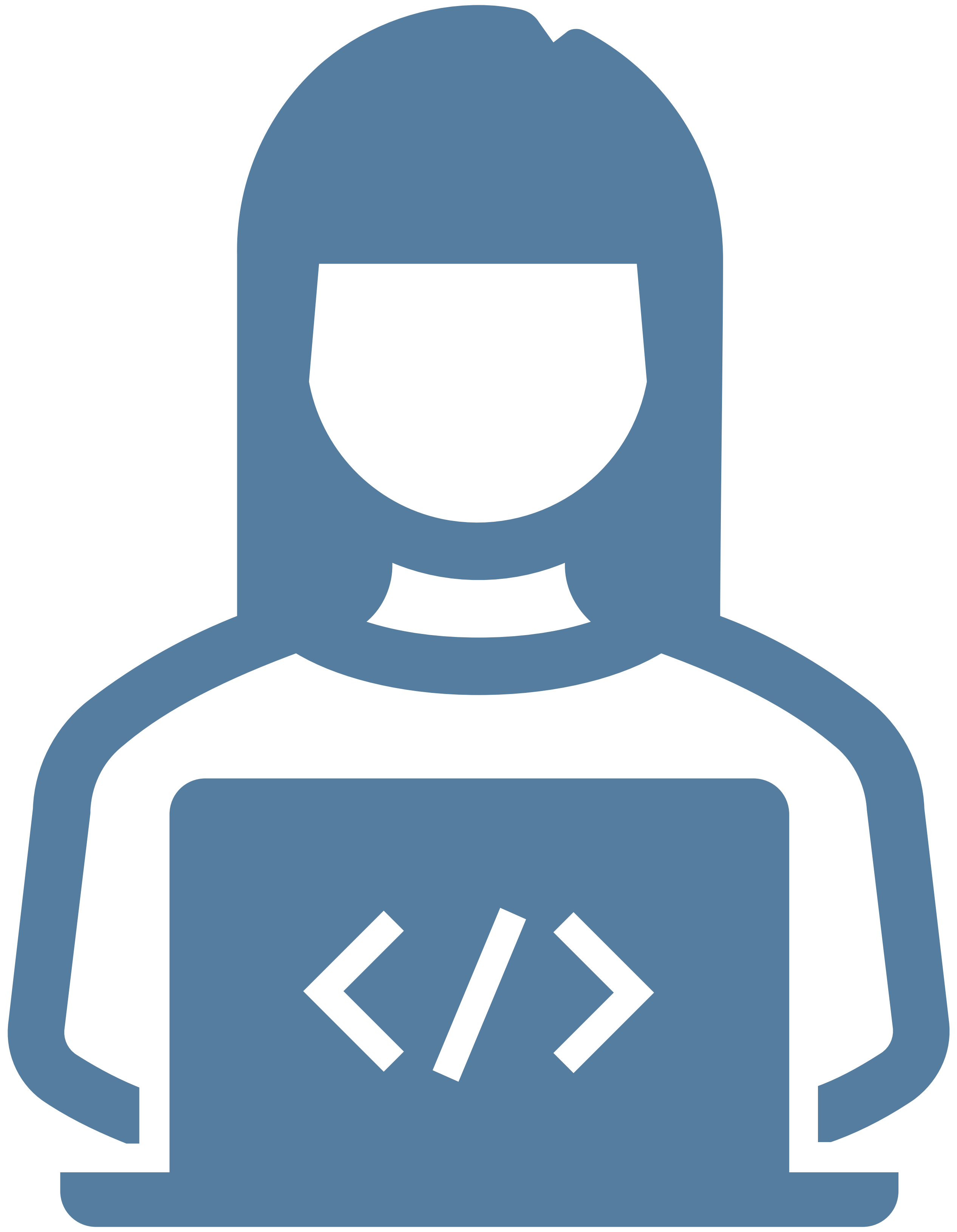
```
# no brain acrobatics

x %>% f(y) %>% g(z)
```

pipe symbol

FROM EXCEL TO R

LIVE CODING 2 – TIDYVERSE



https://www.tidyverse.org/

TIDYVERSE CHEAT SHEET

readr, tidyr, dplyr, …

Read Data (*readr*)

Reading tabular data

There are solutions for multiple data types

```
read_excel()  # using readxl package
read_table()
read_csv()
```

Useful arguments

Skip lines: `read_csv(file, skip=1)`
Read subset: `read_csv(file, n_max=1)`

Data types

readr guesses the types of each column and tells you about it
`("Parsed with column specifications: …")`

Workflow

Tidyverse workflow

```
df %>%
   select(col1)

df %>%
   filter(col1 < x)

df %>%
   summarize(n_1 = n(),
             avg_1 = mean(col),
             sd_1 = sd(col1))
```

Select column

```
df$col1
```

HELP

R Documentation (e.g. enter **`?dplyr::filter`** and see examples)
Much more info and detailed cheat sheets:
https://brianward1428.medium.com/introduction-to-tidyverse-7b3dbf2337d5
It also helps to google "*tidyverse* + *whatever_you_want_to_do*"

Data Manipulation (*dplyr*)

Summary

```
summarize()
count()
```

Group

```
group_by()
```

Functions will manipulate each group separately and combine results.

Extract and sort observations (rows)

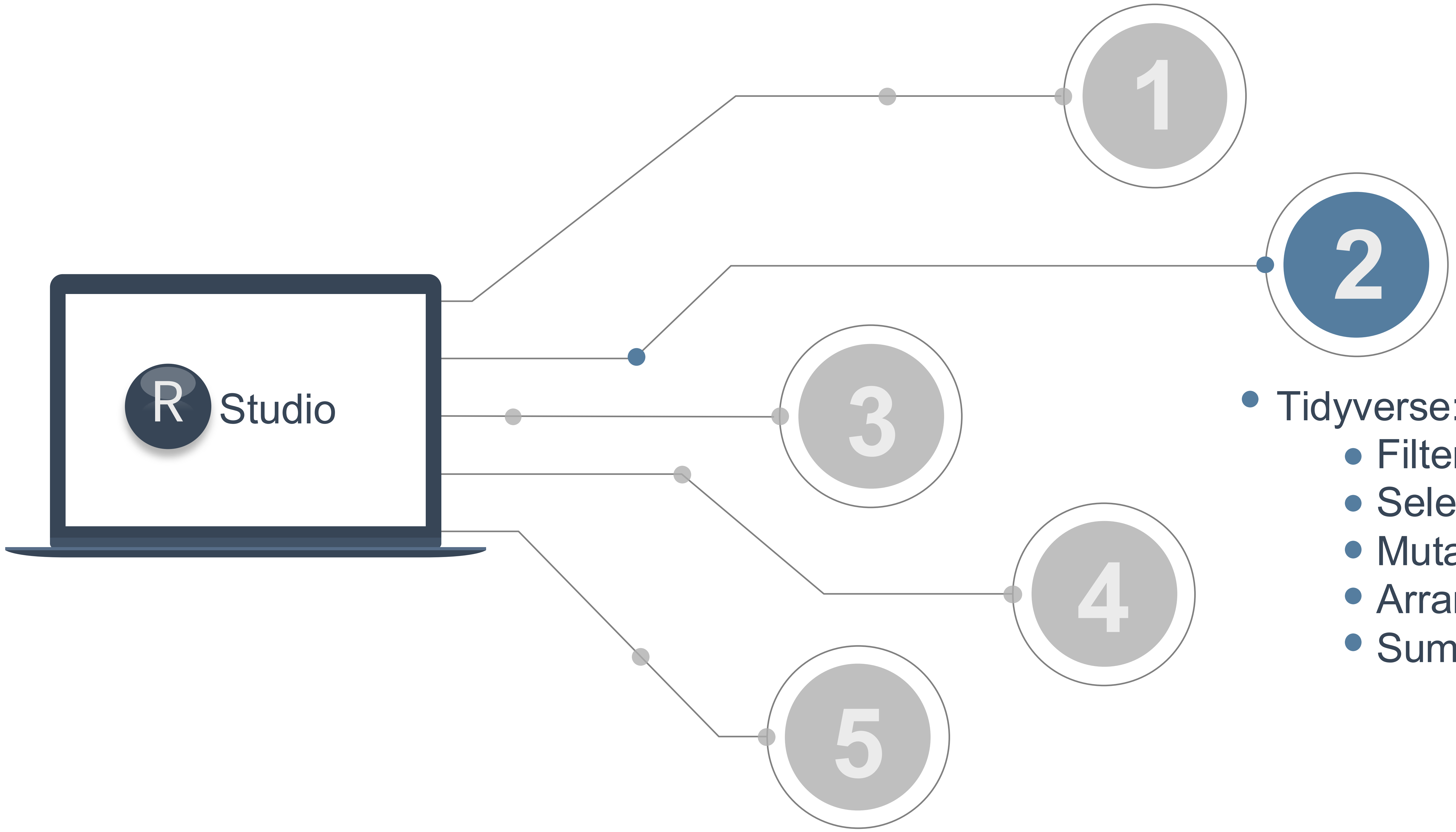
```
filter()  # subset rows by condition
distinct()  # subset to unique values
top_n()  # subset by position
arrange()  # sort low->high, other way with desc()
```

Manipulate variables (columns)

```
select()  # subset rows by condition
mutate(new_name = f(column))
mutate(new_name = ifelse(col1 < x, "Yes", "No"))
mutate(new_name = col1 + col2)
```

TIDYVERSE

EXERCISE 2

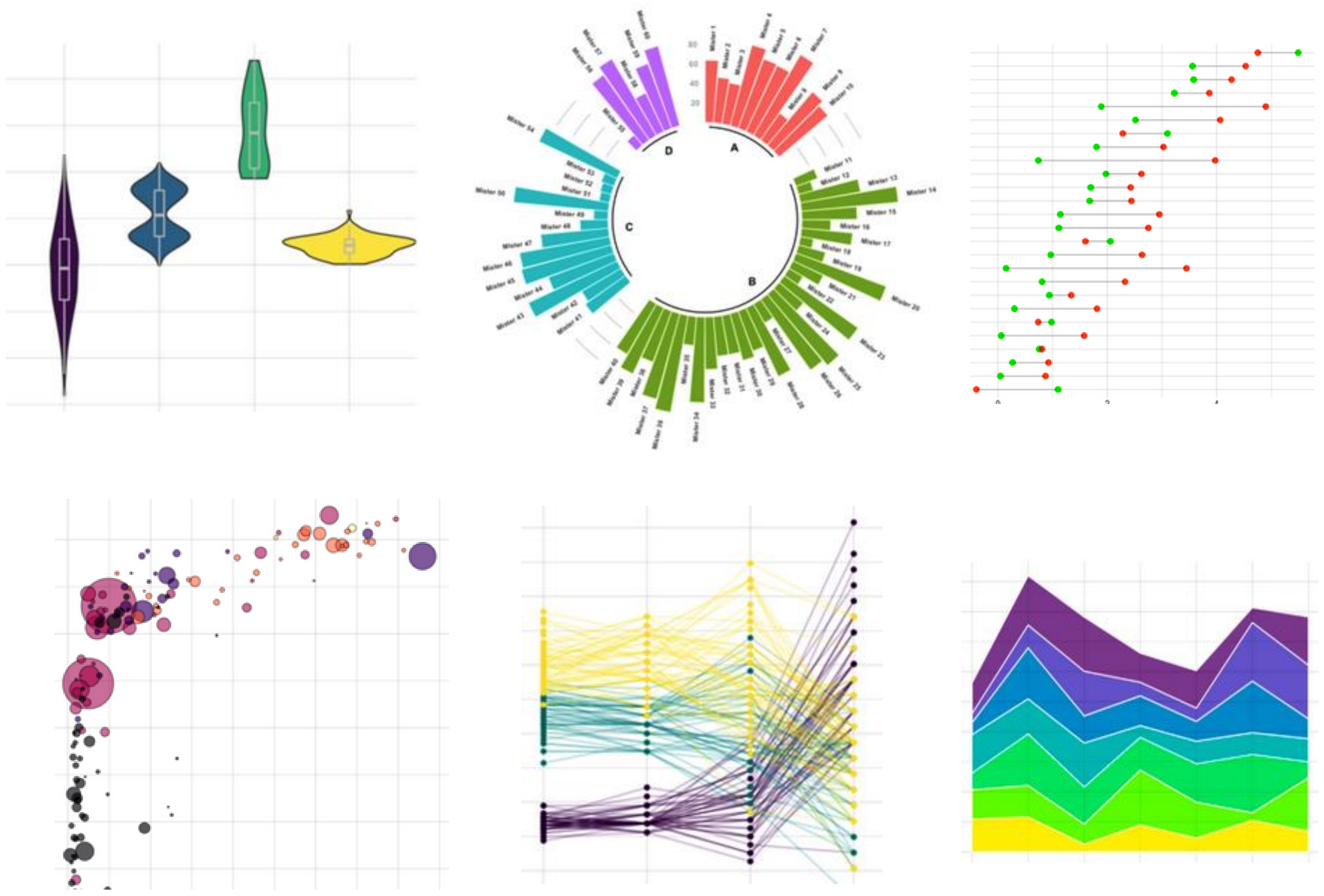


- Tidyverse:
  - Filtering
  - Selecting
  - Mutating
  - Arranging
  - Summarizing

https://www.r-graph-gallery.com/

GGPLOT2 - EASY GRAPHICS

- Aesthetically pleasing graphics.
- Well-defined “additive” (+) structure.
- Integrates perfectly with tidy data.
- Great documentation & community



http://tutorials.iq.harvard.edu/R/Rgraphics/Rgraphics.html

GGPLOT2
ADDITIVE STRUCTURE

DATASET, SAMPLES & OBSERVATIONS

```
ggplot(df,
       aes(x = Sample, y = Measure))
```

DEFINE PLOT TYPE

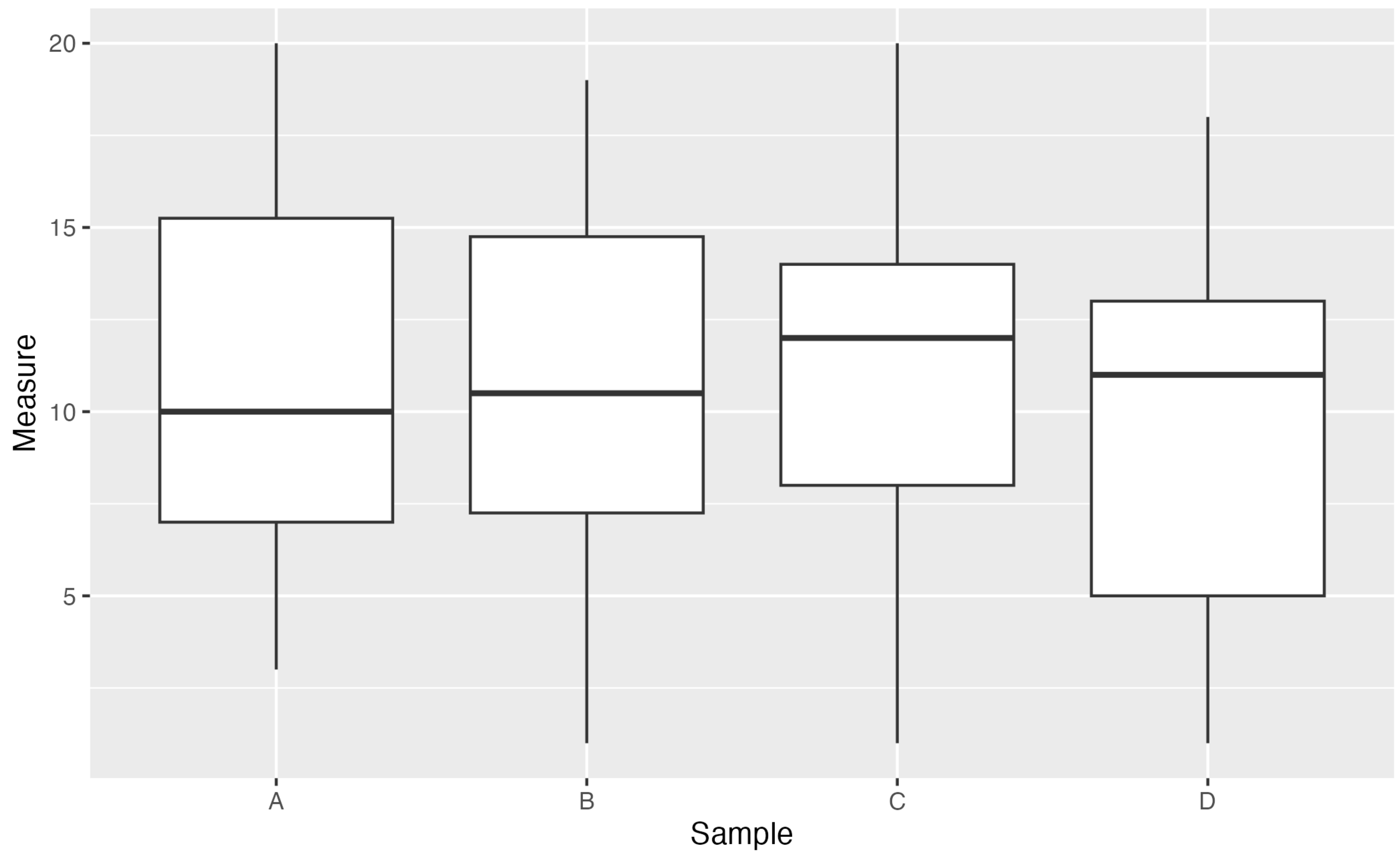
```
ggplot(df,
       aes(x = Sample, y = Measure)) +
  geom_boxplot()
```

```
head(df)
```

Description: df [6 × 2]

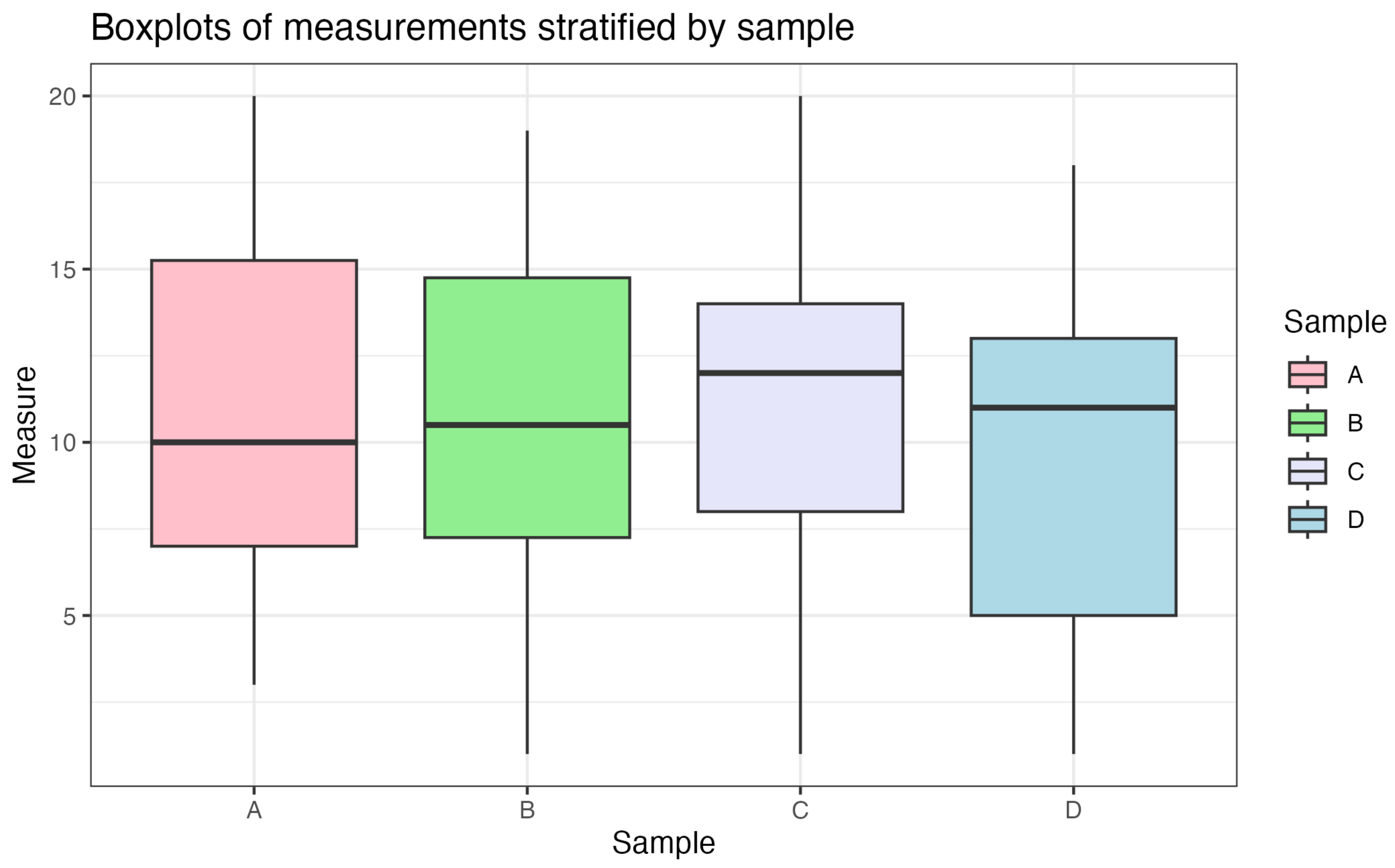
	Sample <chr>	Measure <int>
1	C	20
2	C	14
3	C	3
4	B	8
5	C	16
6	B	12

6 rows



http://tutorials.iq.harvard.edu/R/Rgraphics/Rgraphics.html

GGPLOT2 ADDITIVE STRUCTURE



DATASET, SAMPLES & OBSERVATIONS

```
ggplot(df,
       aes(x = Sample, y = Measure))
```

DEFINE PLOT TYPE

```
ggplot(df,
       aes(x = Sample, y = Measure)) +
  geom_boxplot()
```

COLOR BY GROUP

```
ggplot(df,
       aes(x = Sample, y = Measure,
           fill = Sample)) +
  geom_boxplot()
```

TITLE AND LEGEND

```
… +
  labs(title = ”Boxplots of
measurements stratified by sample”)
```

CUSTOM COLORS

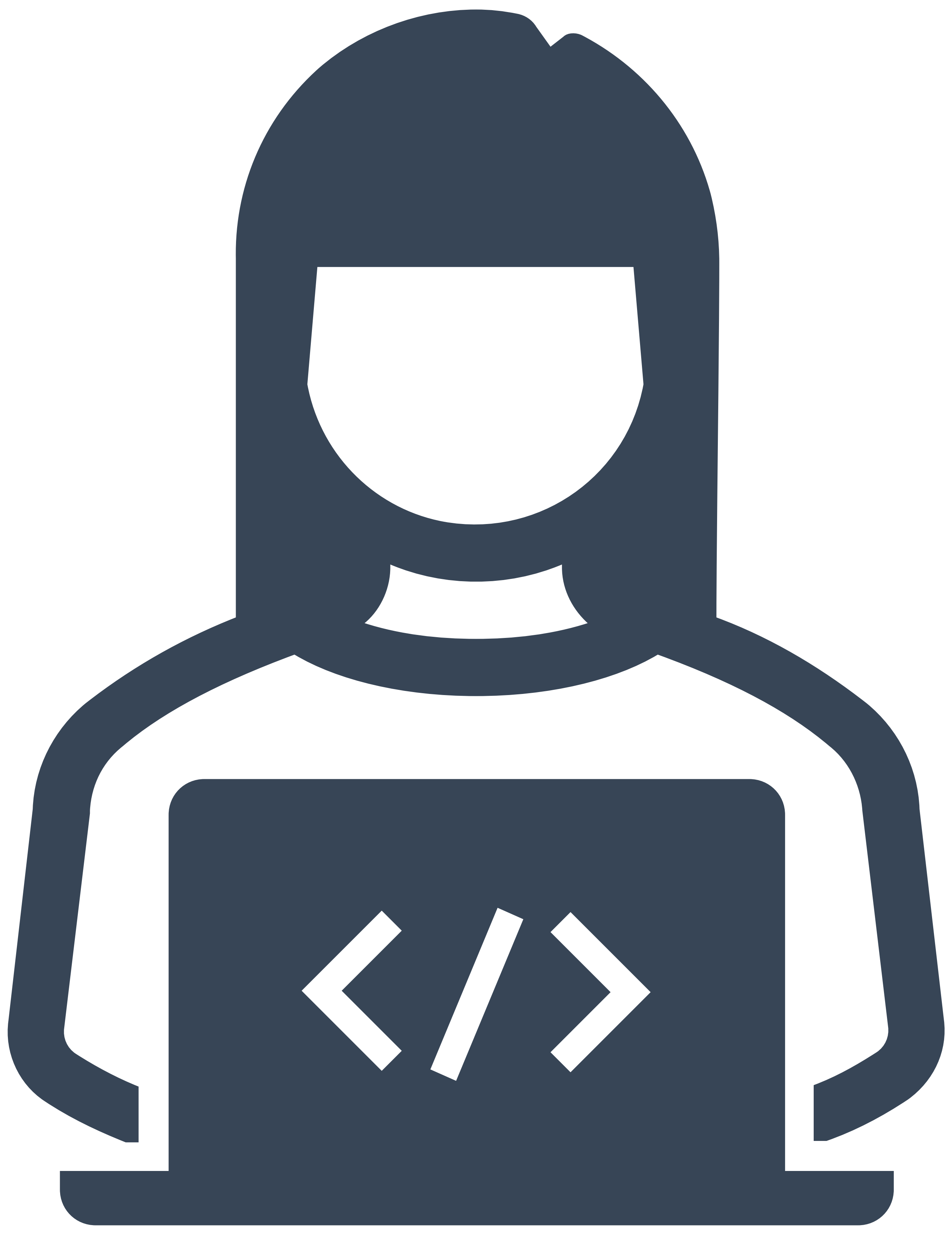
```
… +
  scale_fill_manual(values =
c(”pink”, ”lightgreen”, ”lavender”,
”lightblue”))
```

BACKGROUND

```
… +
  theme_bw()
```

FROM EXCEL TO R

LIVE CODING 3 – GGPLOT 2



https://rstudio.com/wp-content/uploads/2016/11/ggplot2-cheatsheet-2.1.pdf

GGPLOT CHEAT SHEET

GET STARTED

Define Plot:

```
ggplot(data = my.data,
       aes(x = x.var,
           y = y.var))
```

Add Plot Type:

```
… + geom_point() # scatter plot
… + geom_line()
… + geom_boxplot()
… + geom_col()
… + geom_density()
… + geom_histogram()
```

COLORS

One Color:

```
ggplot(…, aes(…),
color = "green")
```

Color Fill by Group:

```
ggplot(…, aes(…,
fill = z.var))
```

Custom Colors:

```
… + scale_*_manual(values = c())
ex: scale_color_manual(values = c("blue", "pink"))
```

COLOR SCALES & THEMES

More Colors:

```
… + scale_fill_grey(start = 0.2, end = 0.8)
… + scale_fill_gradient(low="white", high="red")
```

Labels:

```
… + labs(title = "Title",
         x = "X label",
         y = "Y label")
```

TEXT

Theme:

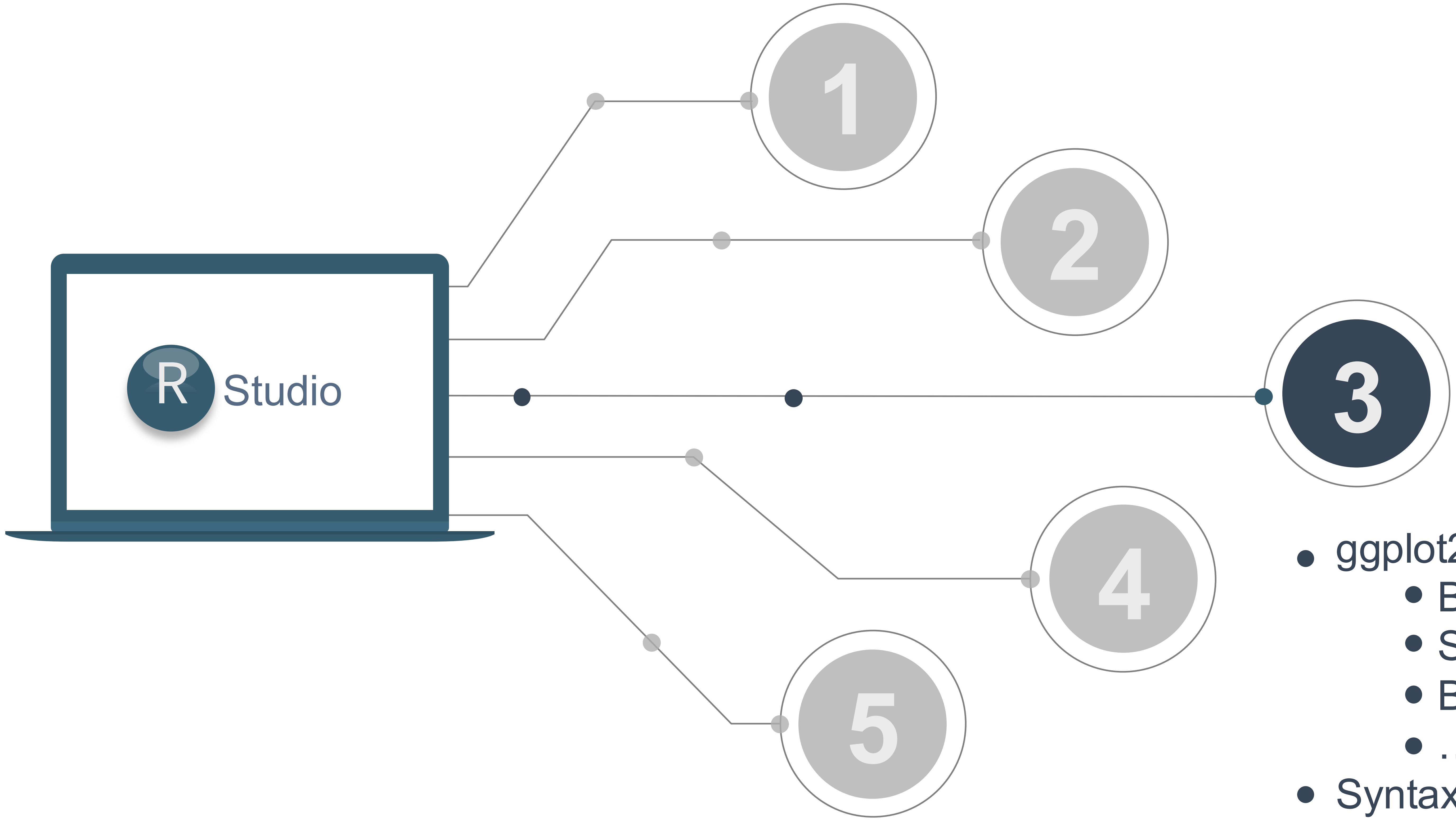
```
… + theme_bw()
… + theme_minimal()
… + theme_dark()
… + theme_classic()
```

Text:

```
… + theme(legend.position = *)
* = "none", "top", "bottom", "left", "rigth"
```

GGPLOT2

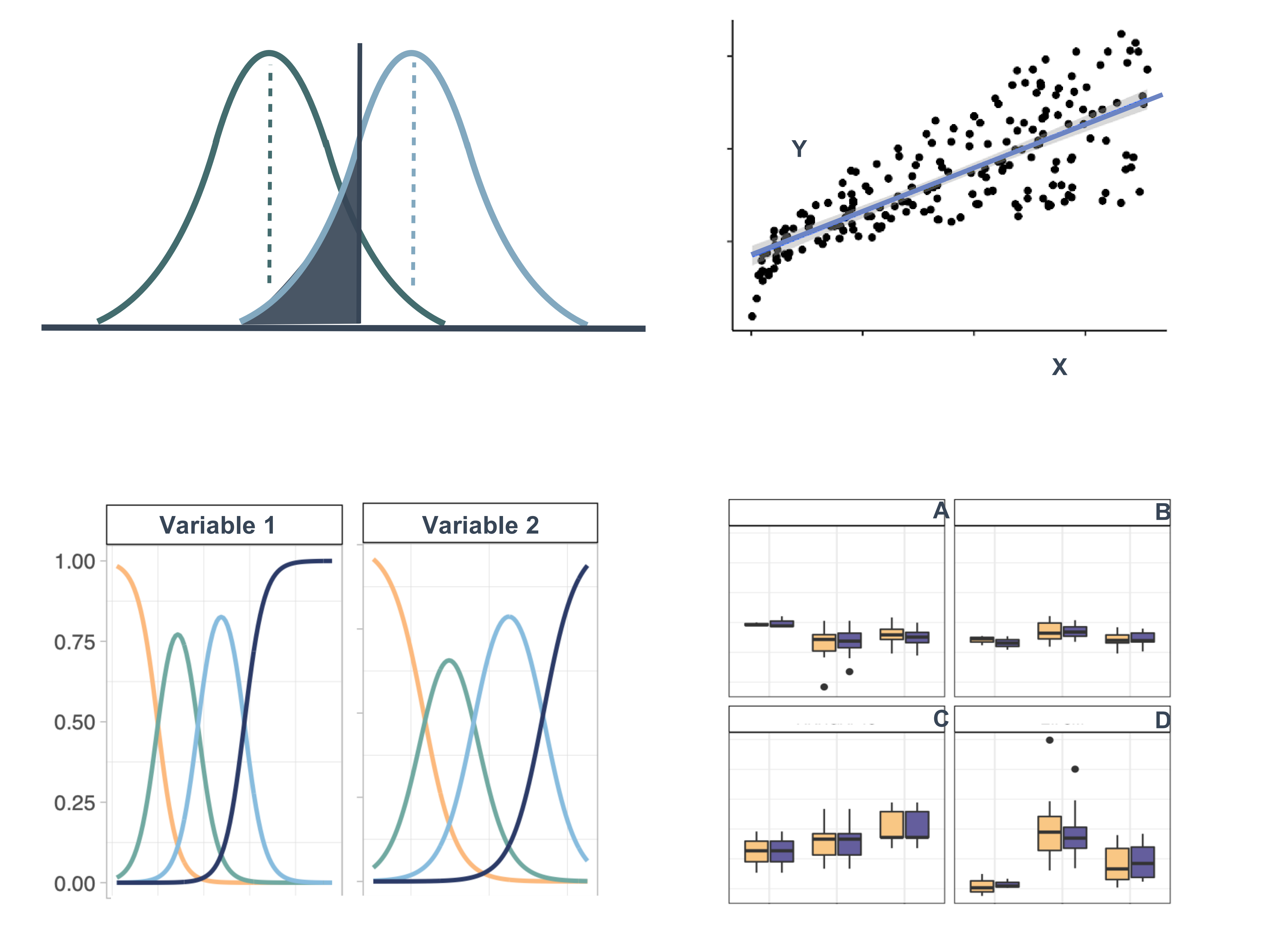
EXERCISE 3



- ggplot2
  - Bar plot
  - Scatter plot
  - Boxplot
  - …
- Syntax
- Aesthetics

http://www.sthda.com/english/

R - A STATISTICAL SCRIPTING LANGUAGE



MODEL FUNCTIONS

`lm(), glm()`
`lmer(), glmer(),`
`nls(), …`

EMMEANS PACKAGE

`emmeans(),`
`pairs(), cld()`

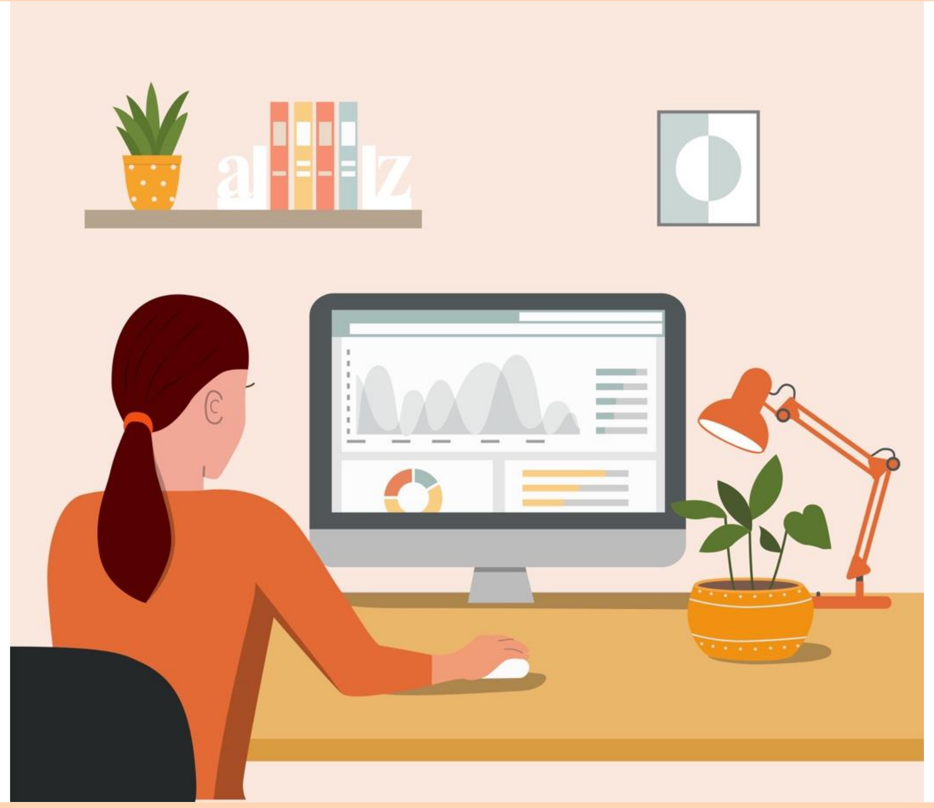
APPLY TO MODEL

`summary(), anova(),`
`confint(), predict(),`
`drop1(), update(),`
`step(), …`

MORE FUNCTIONS

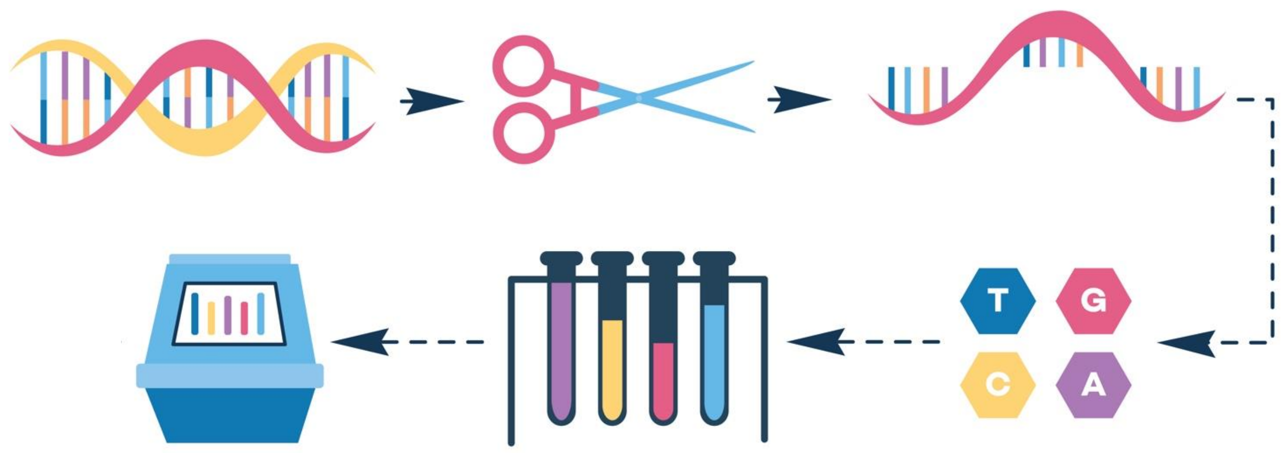
`t.test(), cor(),`
`cor.test(), aov(),`
`quantile(),`
`p.adjust(),`
`rank(), …`

Let’s use R in a statistical analysis



YOU

The researcher with R skills!



YOUR DATA

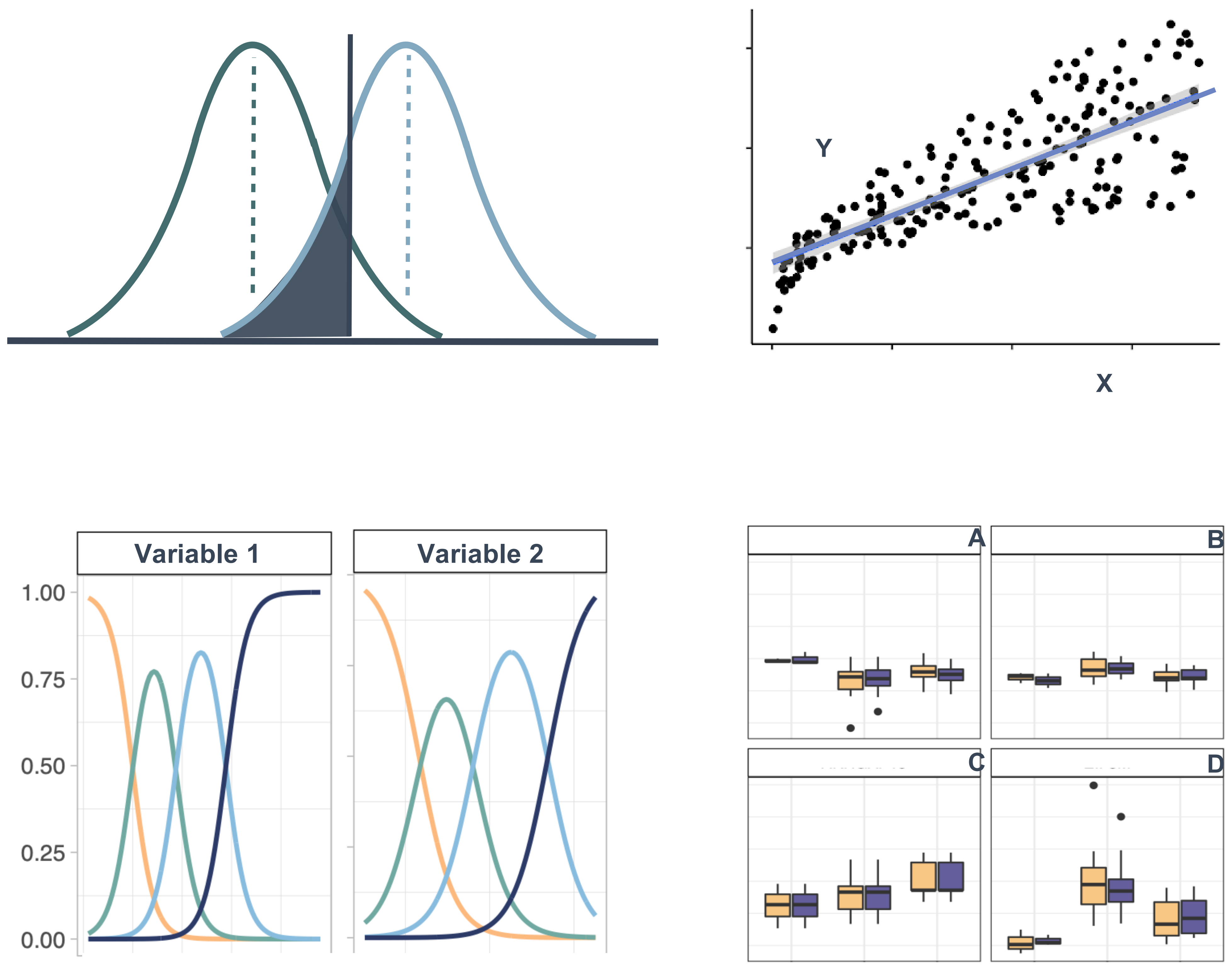
Gene expression data, i.e. how much a gene is used to produce a protein. Multiple genes measured in one experiment.

YOUR DISEASE OF INTEREST

Psoriasis, an immune-mediated skin disorder.

http://www.sthda.com/english/

R - A STATISTICAL SCRIPTING LANGUAGE

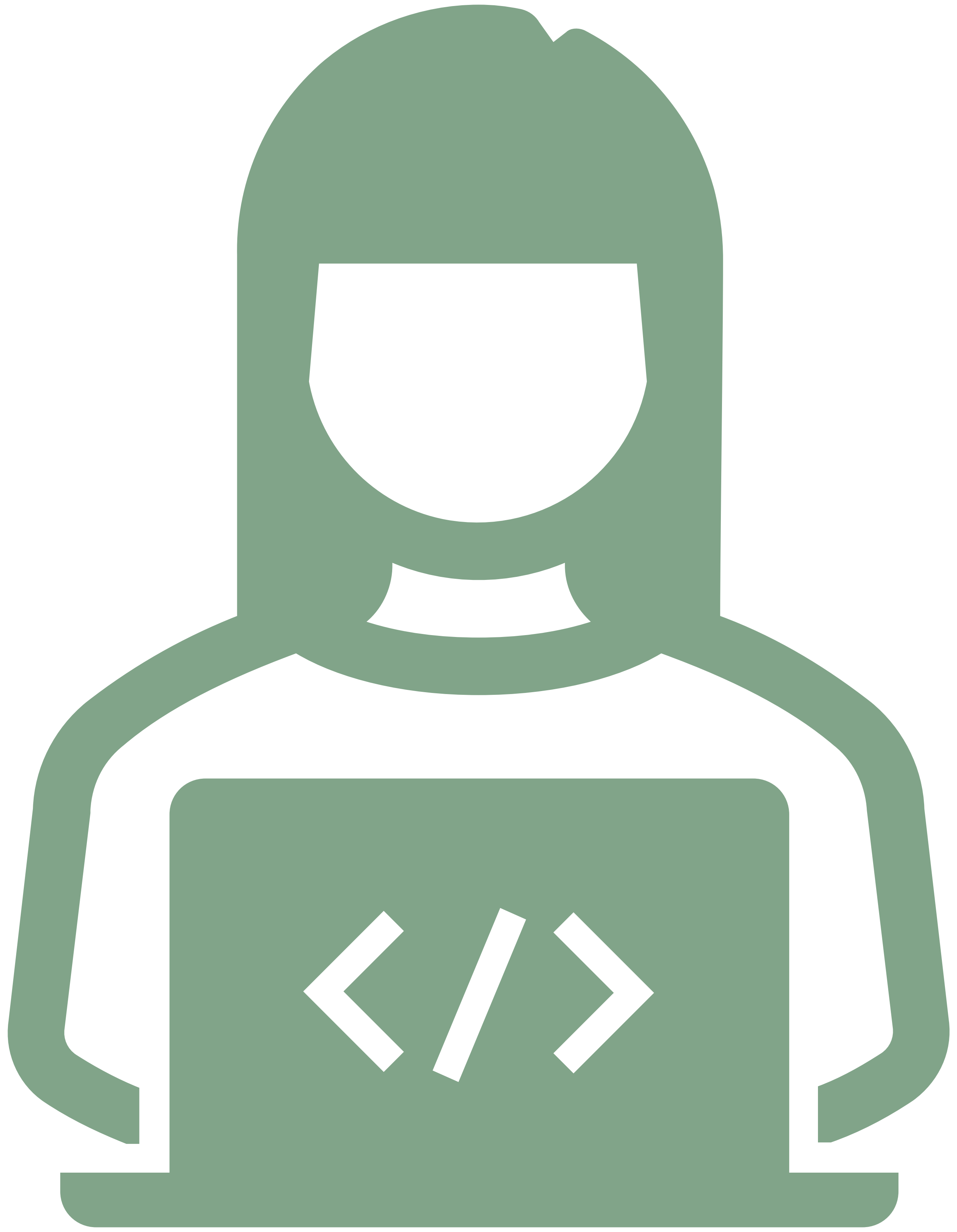


During this session:

- *Cooperatively* discuss and share ideas about the data
- Apply steps of basic statistical analysis for hypothesis testing consistent with the given data
- Suggest conclusions based on your analysis, regarding the association between psoriasis and gene expression levels

FROM EXCEL TO R

LIVE CODING 4 – APPLIED STATS



https://rstudio.com/resources/cheatsheets/ for various relevant cheat sheets.
Other example: https://www.dummies.com/programming/r/statistical-analysis-with-r-for-dummies-cheat-sheet/

STATS CHEAT SHEET

GET STARTED

Import Data:

```
read_excel("my.data.xlsx")
```

Overview of Data:

```
summary(my.data)
nrow(my.data)
length(my.data)
names(my.data)
```

REGRESSION MODELS

Linear:

```
lm(y~x, data=my.data)
confint(model)
```

Logistic:

```
glm(y~x,
data=my.data)
```

Linear Mixed:

```
lmer(y~x + (1|z),
data=my.data)
```

Check Model:

```
summary(model)
par(mfrow=c(2,2))
plot(model)
```

TESTS/ COMPARISONS

ANOVA:

```
anova(model2, model1)
```

F-Test:

```
drop1(model, test="F")
```

Emmeans:

```
emmeans(model,~x)
pairs(emmeans(model,~x))
```

VARIABLES

Check Type:

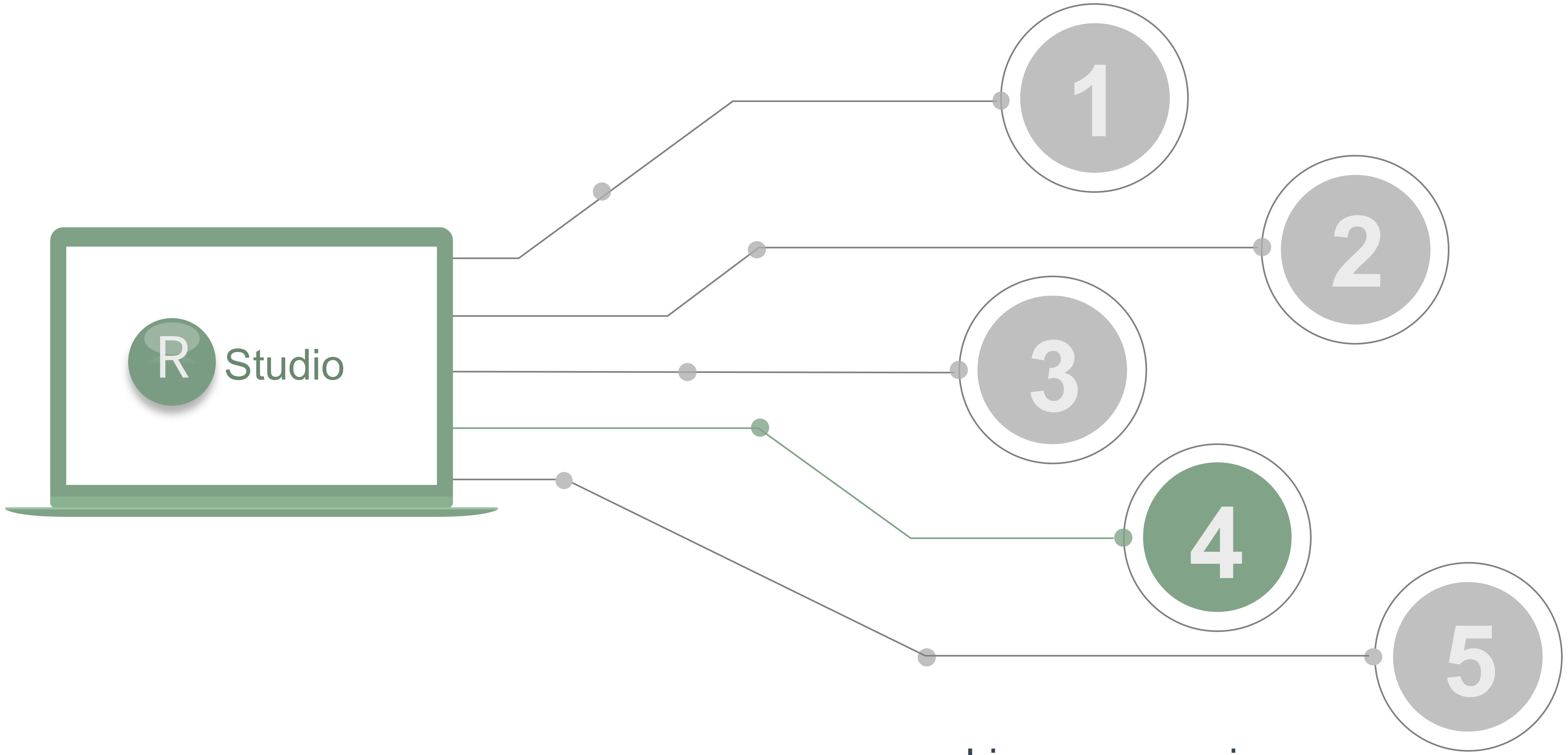
```
table(my.data$x)
is.numeric(my.data$x)
is.factor(my.data$x)
```

Change Type:

```
my.data <- mutate(my.data, x = factor(x))
my.data$z <- as.numeric(my.data$z)
```

Statistics in R

EXERCISE 4

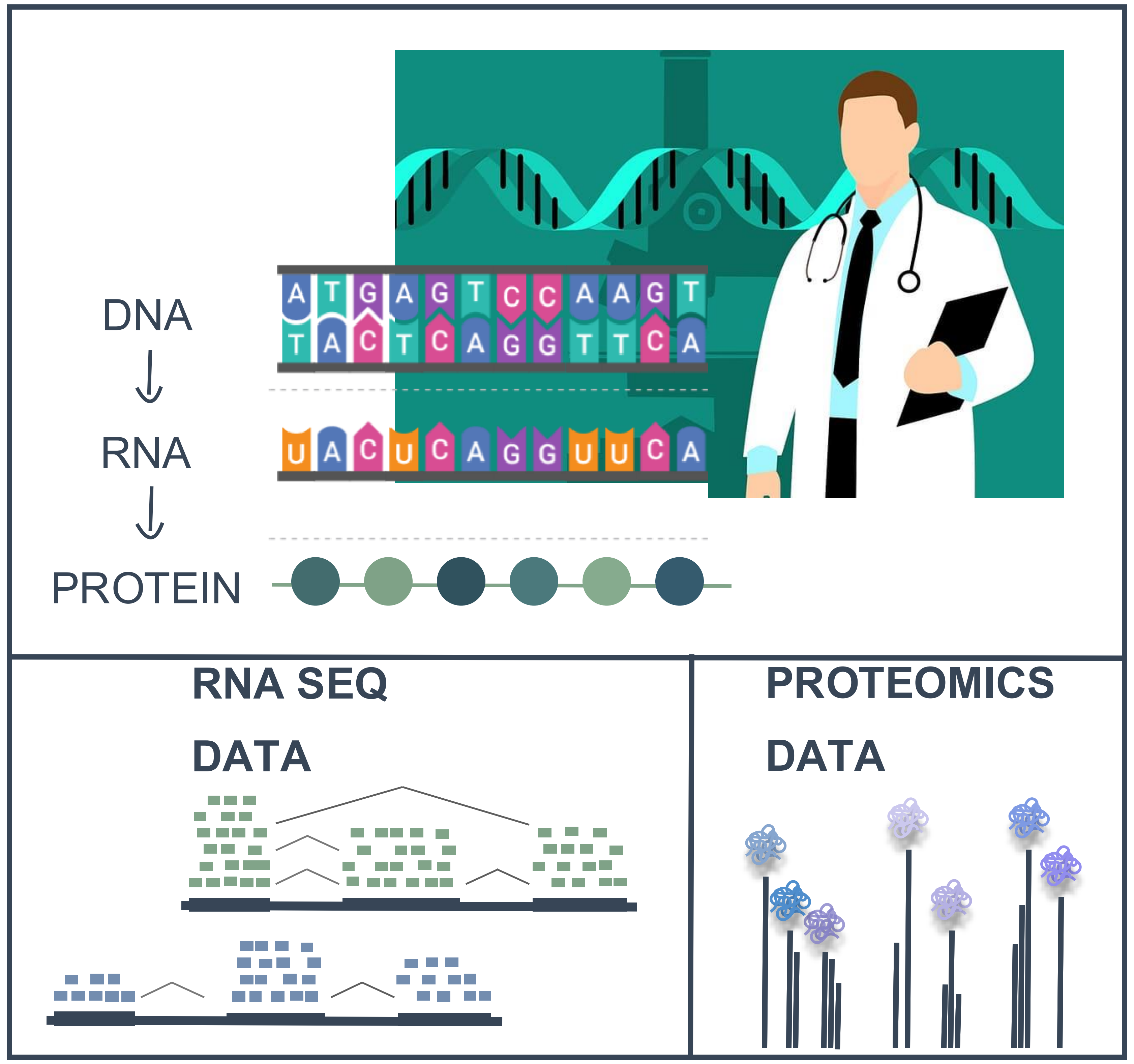


- Linear regression
- Summary Statistics
- ANOVA
- Logistic regression
- Clustering
- Correlation

https://www.r-graph-gallery.com/

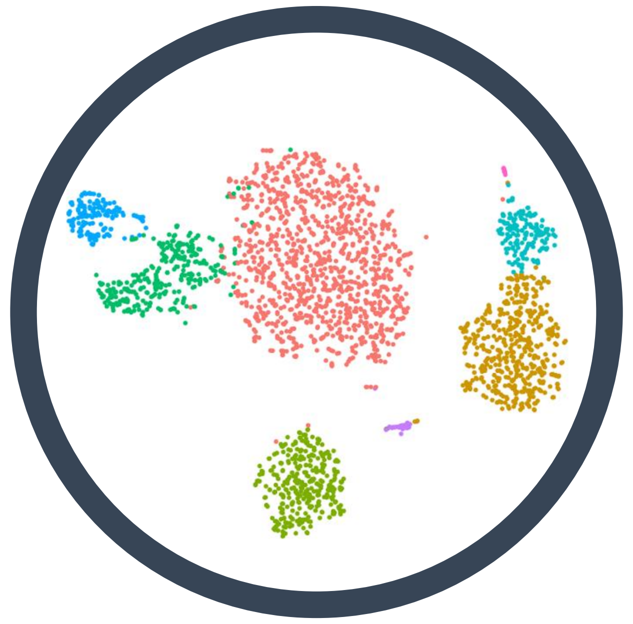
BIOINFORMATICS IN **R**

HIGH THROUGHPUT DATA

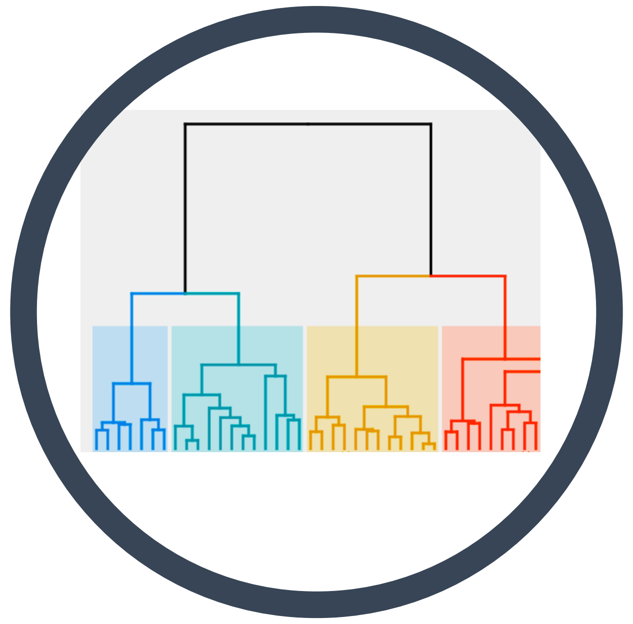


BIOINFORMATIC ANALYSIS

DIMENSIONALITY REDUCTION

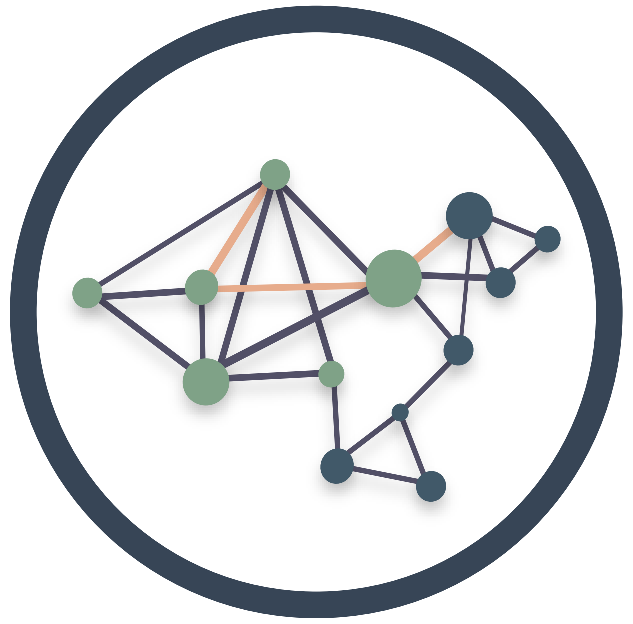


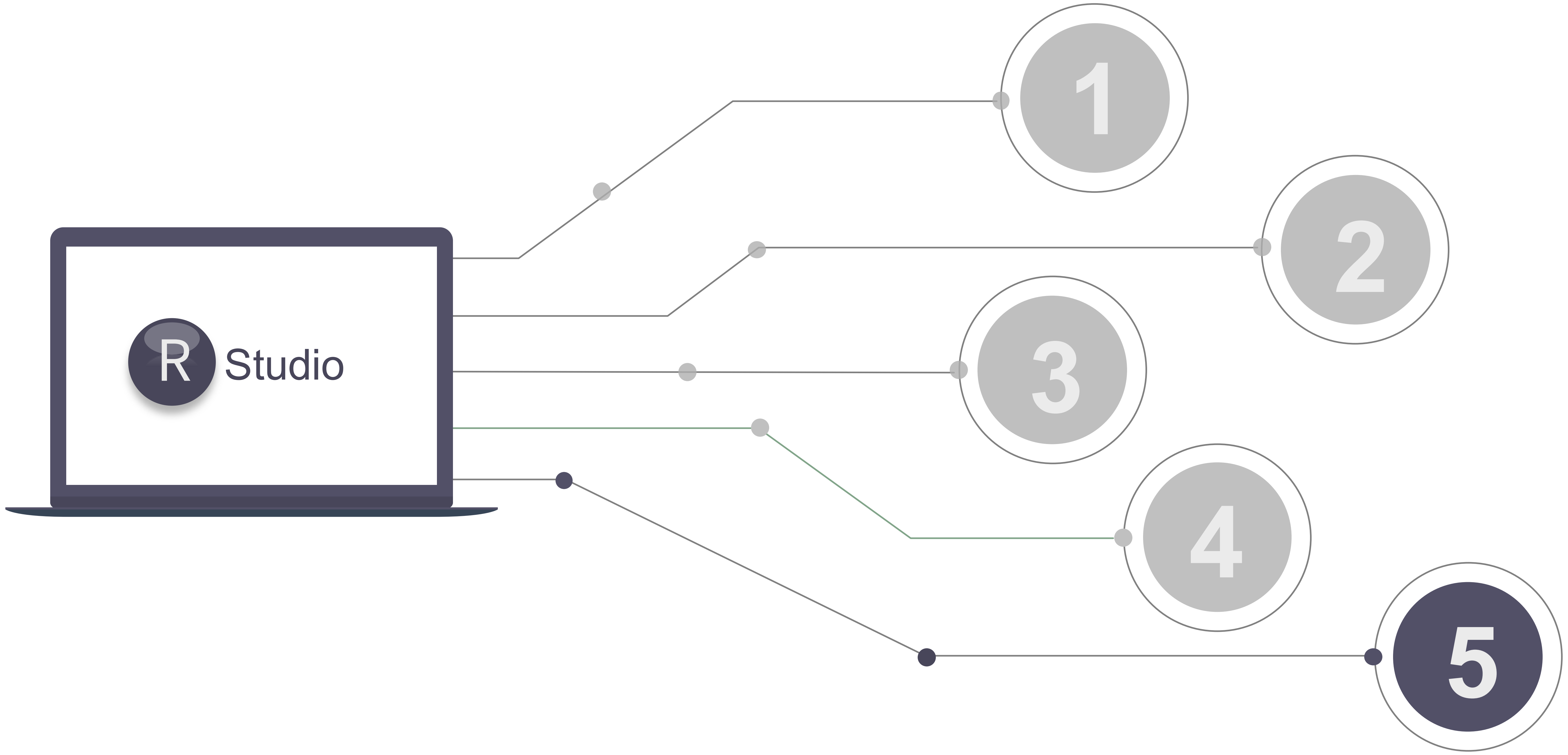
CLUSTERING



MACHINE LEARNING

NETWORK ANALYSIS





- Tidyverse
- ggplot
- Stats
- Quarto

Statistics in R

EXERCISE 5

THE TOP OF THE R ICEBERG

STATISTICAL ANALYSIS

Statistical models (linear, generalized, mixed, …)

Statistical tests (t-test, chisq, anova, …)

Survival analysis (Cox, Kaplan meier)

DATA MANGEMENT

EASY PLOTTING

BIOINFORMATIC ANALYSIS

WANT MO-R-E?

The Section for Biostatistics offers a number statistics-oriented R courses:

https://publichealth.ku.dk/about-the-department/biostat/

Spring

Basic Statistics for Health Researchers (Danish course)
ECTS: 9,0

Epidemiological methods in medical research
ECTS: 7

Advanced topics in health research B
ECTS: 2,8

Statistical methods in bioinformatics
ECTS: 3,5

Statistical analysis of survival data
ECTS: 4,9

Programming and statistical modelling in R
ECTS: 1,6

Bayesian methods in biomedical research
ECTS: 2,4

Psychometric validation of patient reported outcome measures
ECTS: 2,5

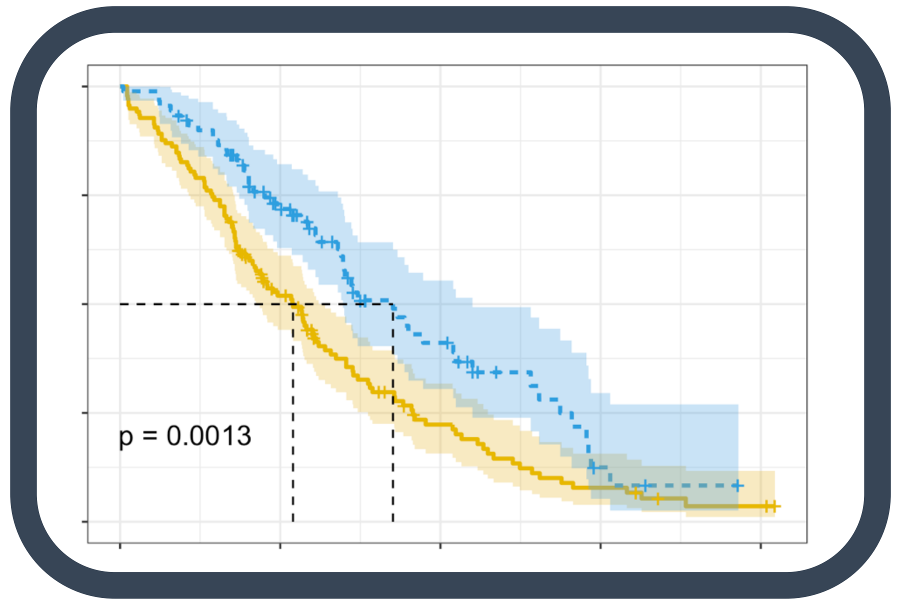
Introduction to validation of patient reported outcome measures
Causal inference I
ECTS: 2,5

Use of the statistical software R
ECTS: 2,1

* These are screenshots. Go to the website and scroll down to ‘Teaching’

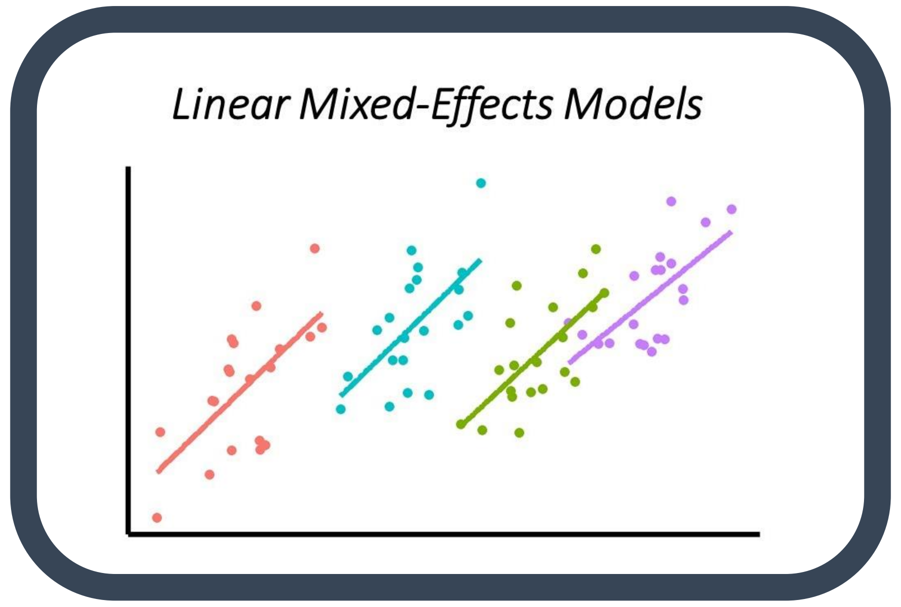
TEASER

STATISTICS in R



Survival Analysis

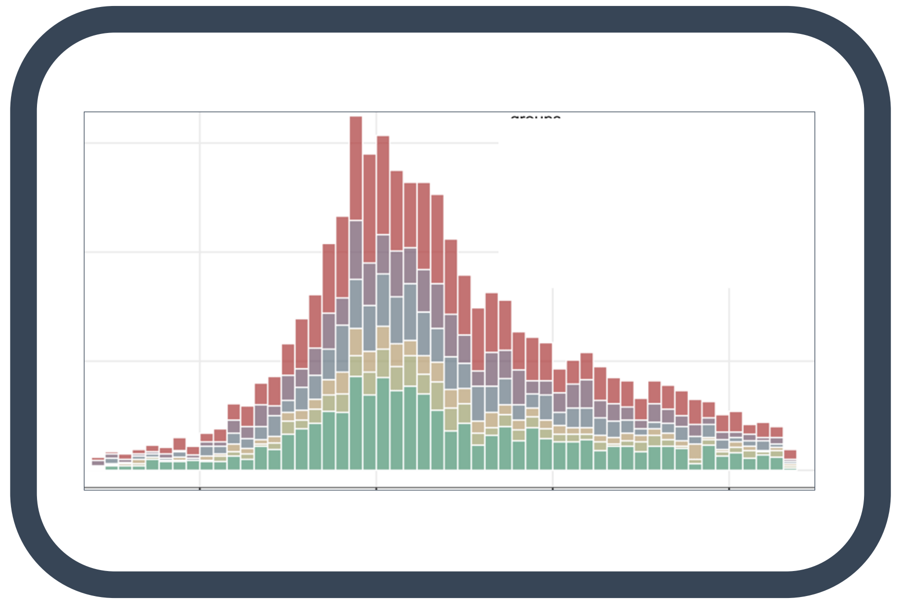
survival: https://rviews.rstudio.com/2017/09/25/survival-analysis-with-r/
survminer: https://cran.r-project.org/web/packages/survminer/survminer.pdf
(https://rpkgs.datanovia.com/survminer/)



Mixed-Effects Models

lme4: https://cran.r-project.org/web/packages/lme4/vignettes/lmer.pdf

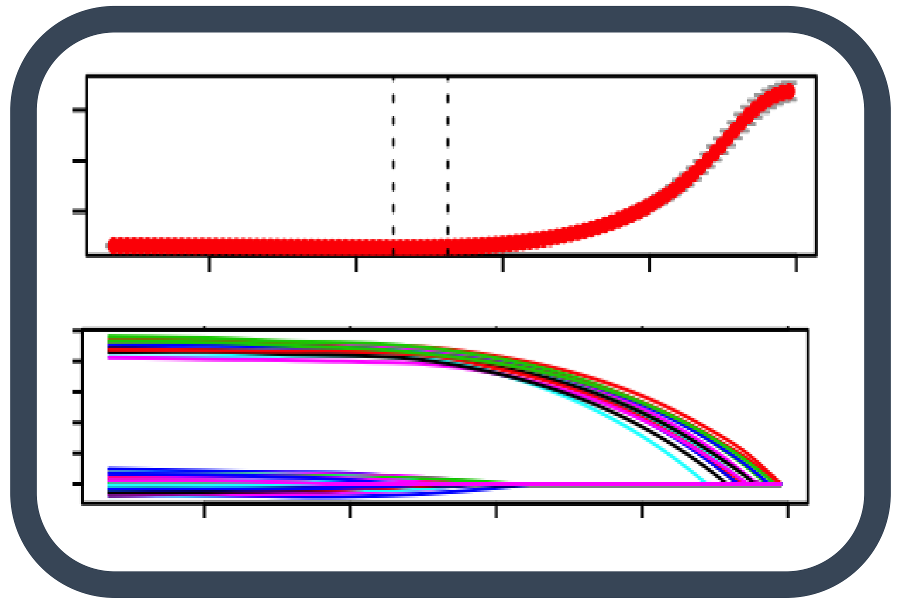
https://cran.microsoft.com/snapshot/2017-08-01/web/packages/sjPlot/vignettes/sjplmer.html
glmmTMB: https://cran.r-project.org/web/packages/glmmTMB/index.html



Epidemiological Analysis

Epi: https://cran.r-project.org/web/packages/Epi/index.html

pubh: https://rviews.rstudio.com/2020/03/05/covid-19-epidemiology-with-r/
https://cran.r-project.org/web/packages/incidence/vignettes/customize_plot.html
https://rviews.rstudio.com/2020/03/05/covid-19-epidemiology-with-r/



Elastic-Net Regression

(R
glmnet: https://cran.r-project.org/web/packages/glmnet/glmnet.pdf

elasticnet: https://cran.r-project.org/web/packages/elasticnet/elasticnet.pdf

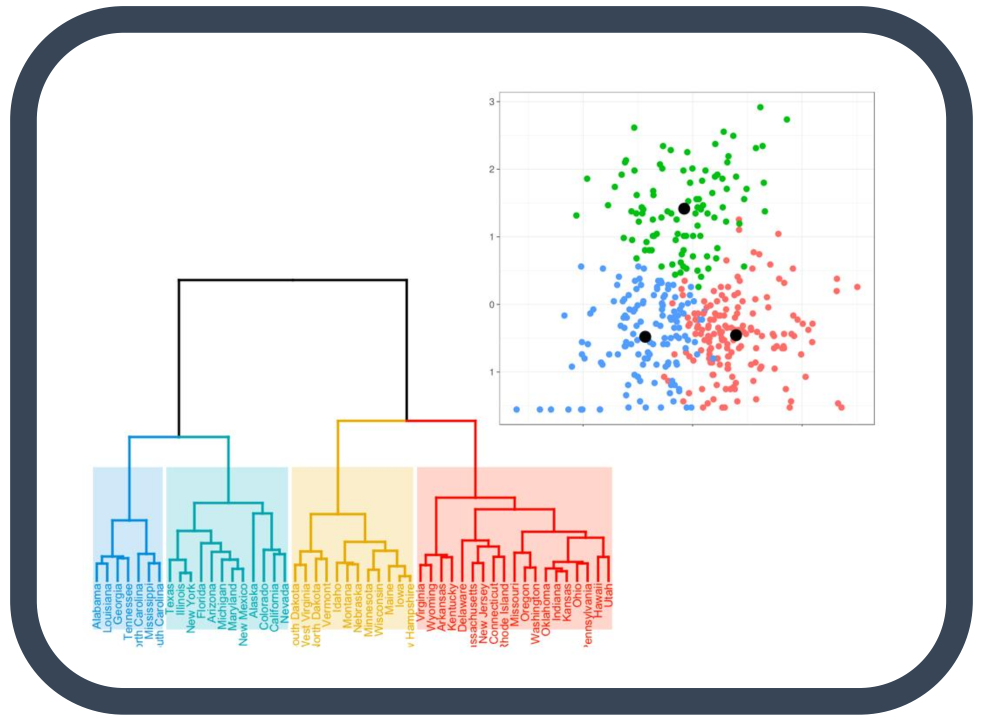
https://www.datacamp.com/community/tutorials/tutorial-ridge-lasso-elastic-net

TEASER

Machine Learning

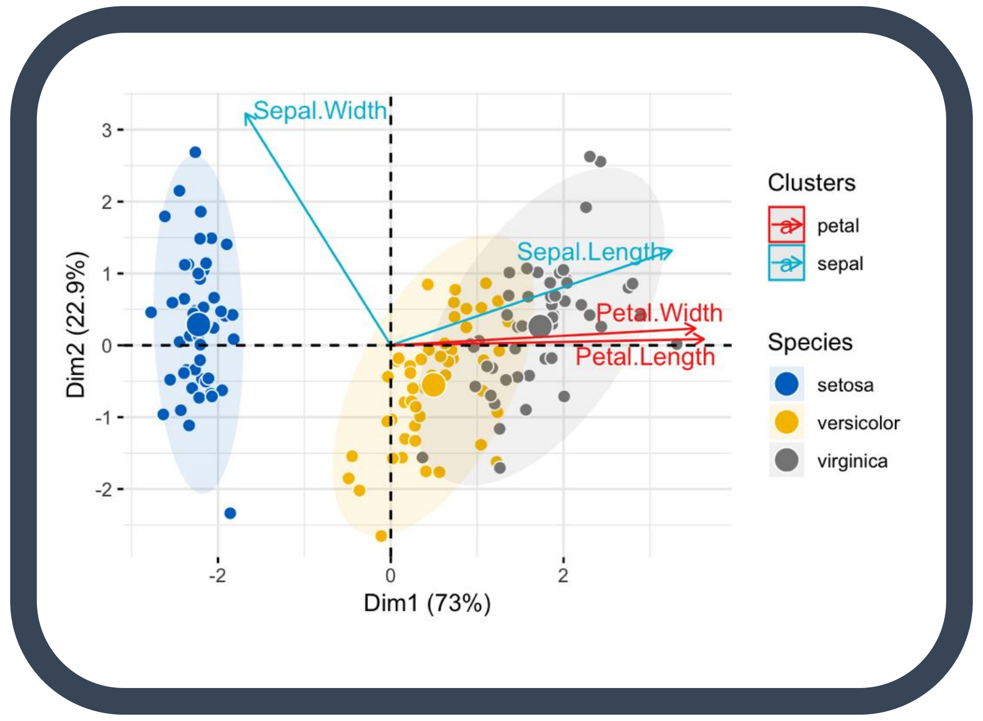
https://lgatto.github.io/IntroMachineLearningWithR/an-introduction-to-machine-learning-with-r.html

Clustering



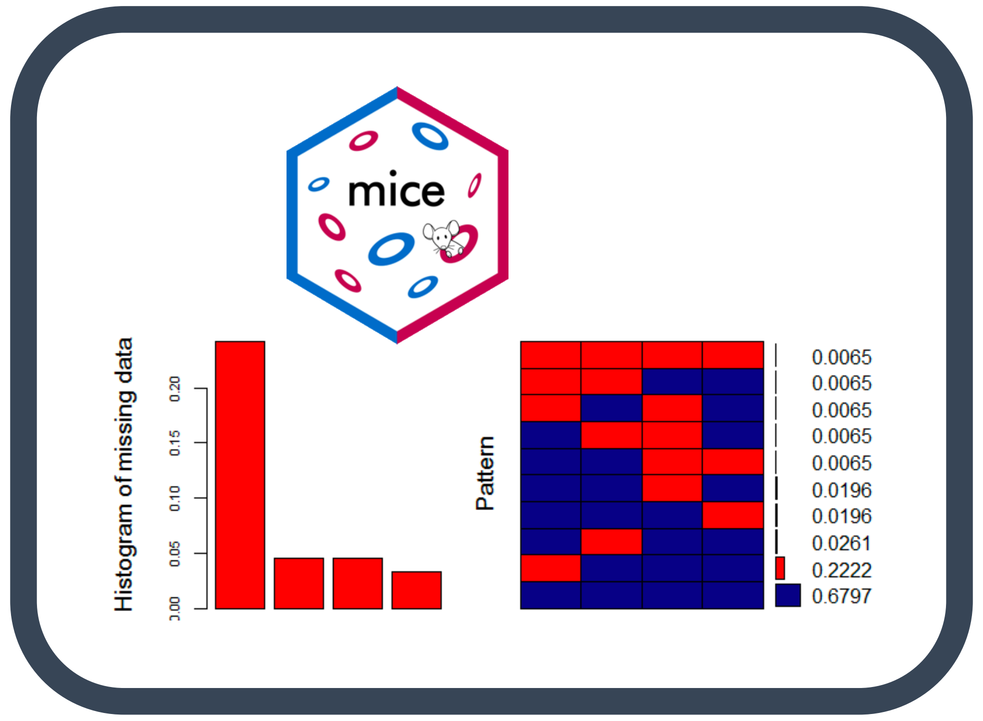
https://statsandr.com/blog/clustering-analysis-k-means-and-hierarchical-clustering-by-hand-and-in-r/

Feature Selection: PCA



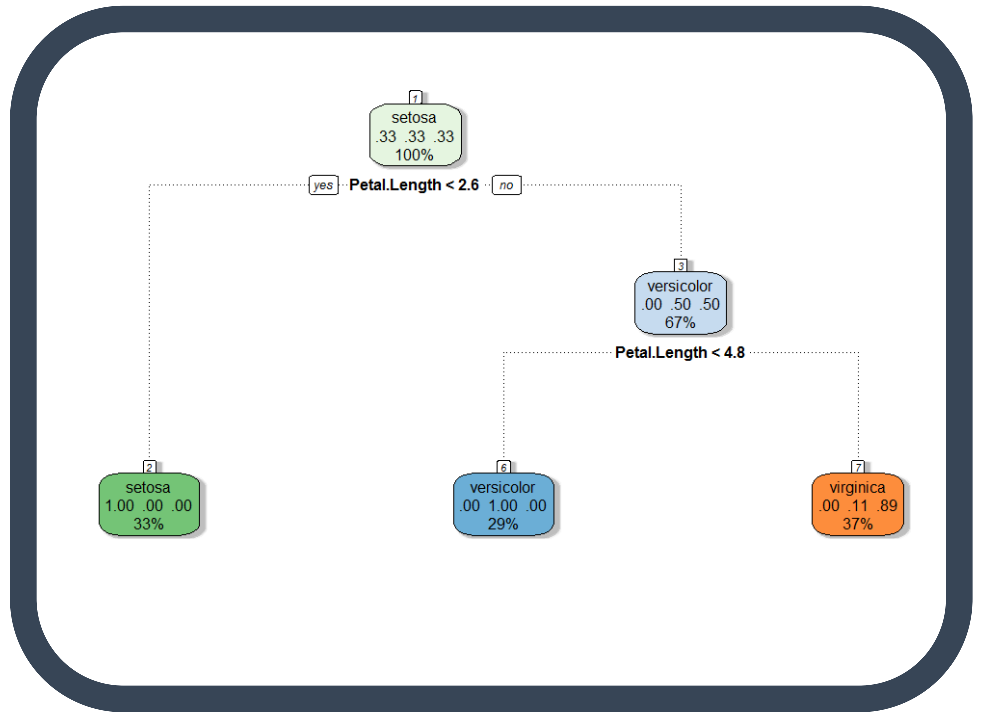
https://bioconductor.org/packages/release/bioc/vignettes/PCAtools/inst/doc/PCAtools.html

Missing Data



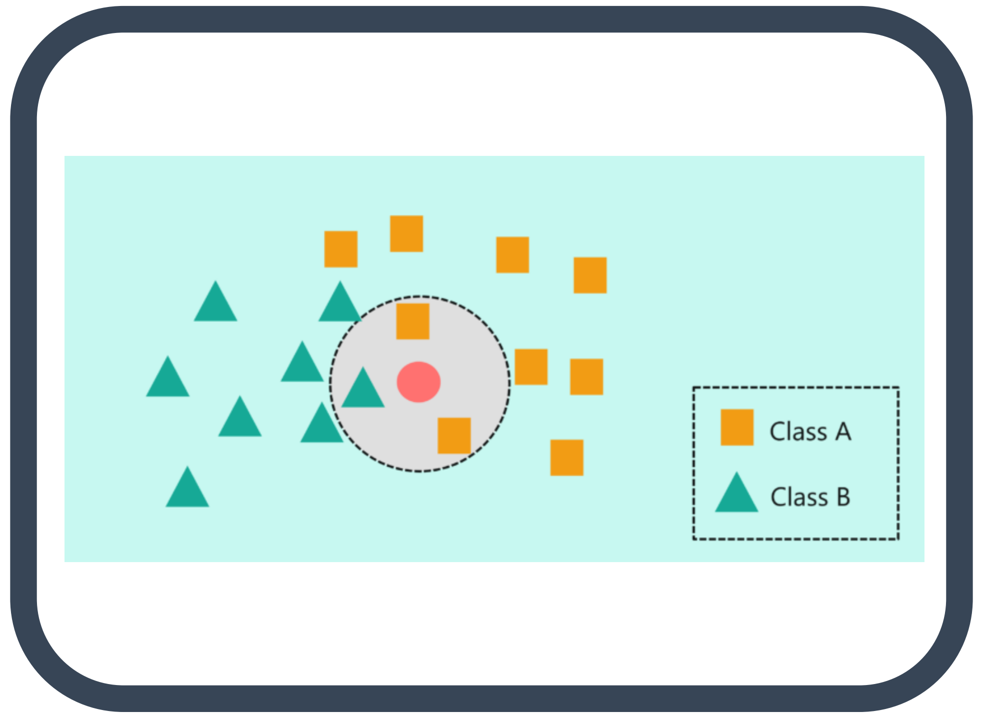
https://amices.org/mice/
https://datascienceplus.com/imputing-missing-data-with-r-mice-package/

Random Forest



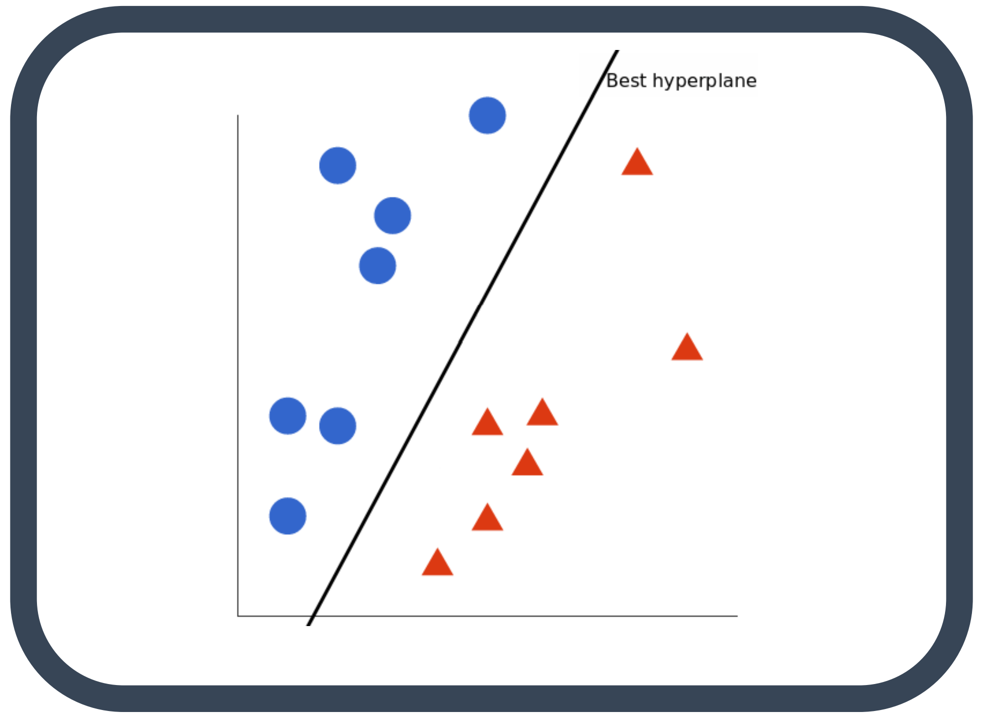
https://www.blopig.com/blog/2017/04/a-very-basic-introduction-to-random-forests-using-r/

kNN



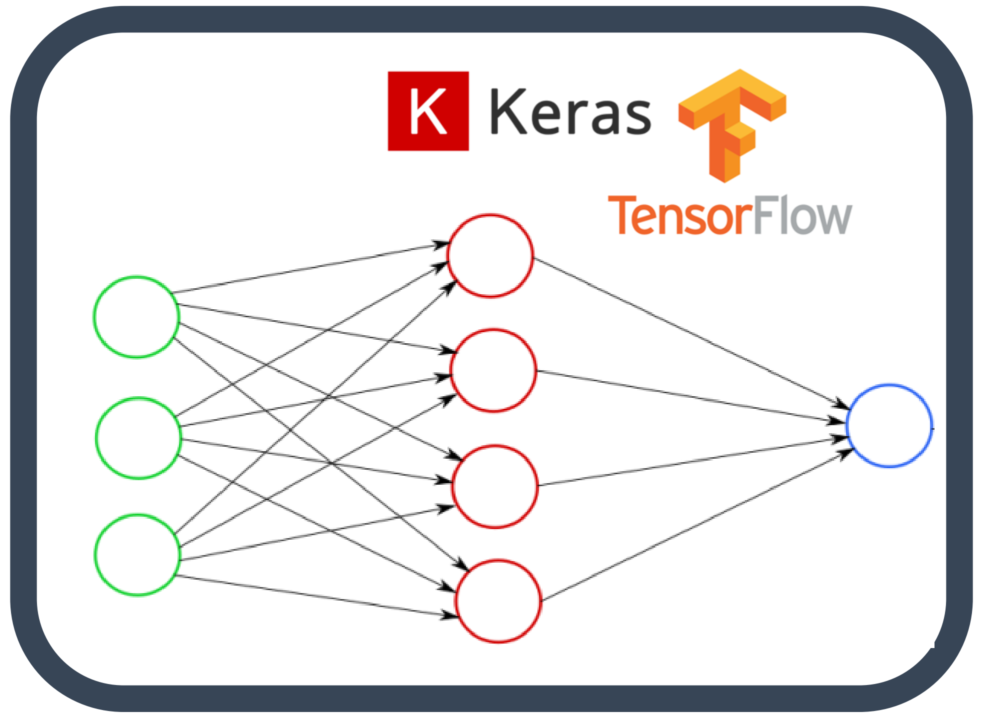
https://www.edureka.co/blog/knn-algorithm-in-r/

SVM



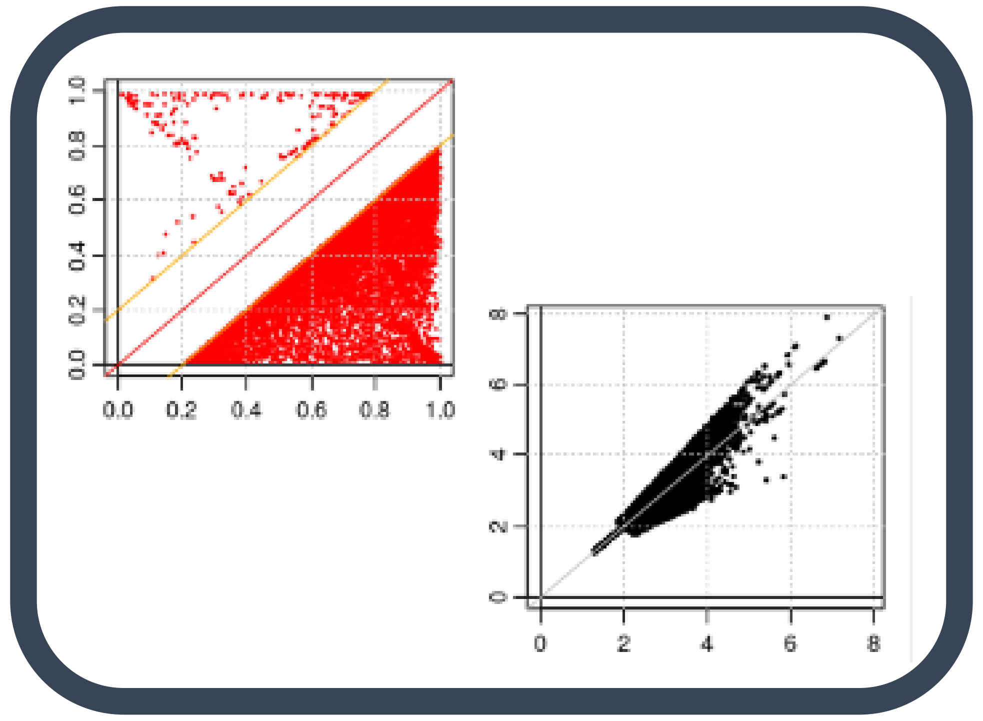
https://cran.r-project.org/web/packages/e1071/vignettes/svmdoc.pdf

Neural Networks



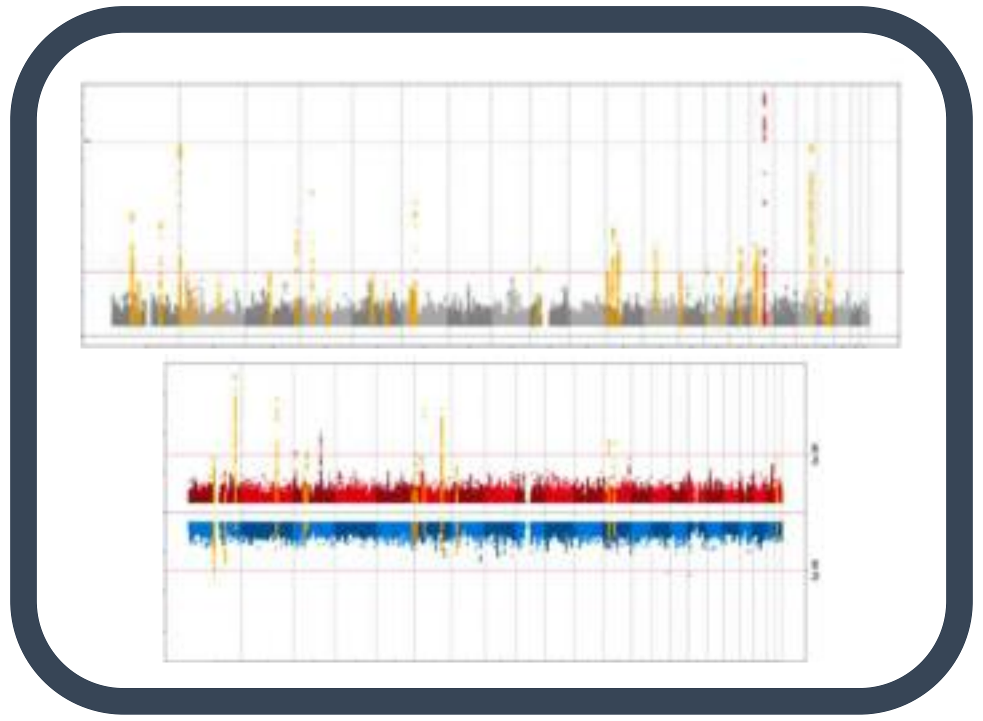
https://keras.rstudio.com/
https://tensorflow.rstudio.com/

GWAS - QC & Data Harmonization



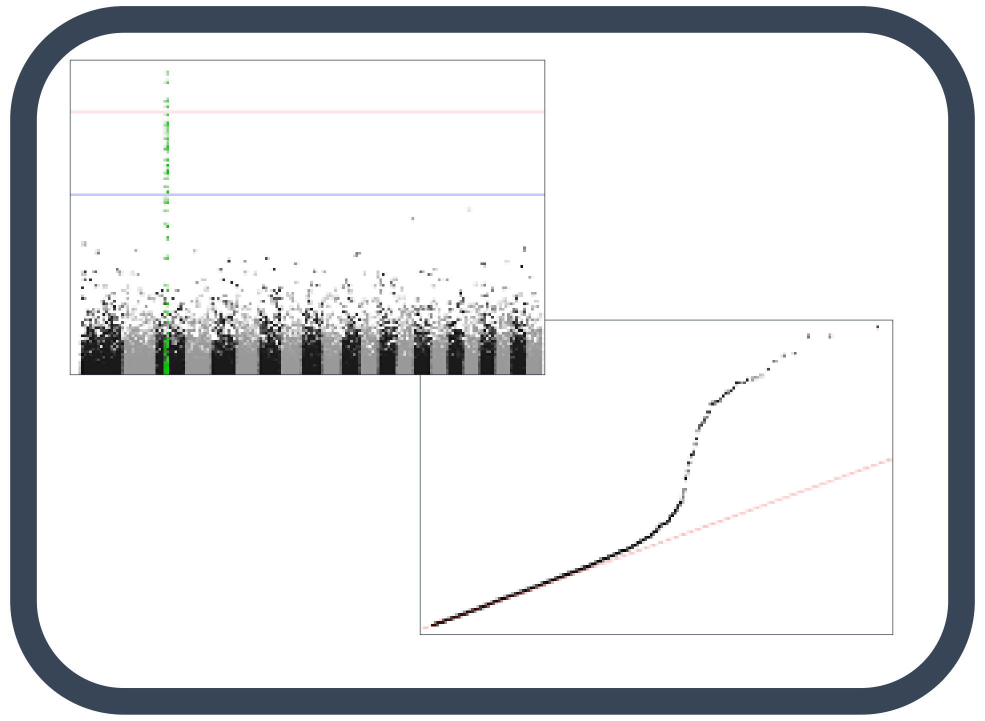
EasyQC: https://www.uni-regensburg.de/medizin/epidemiologie-praeventivmedizin/genetische-epidemiologie/software/

GWAS Data Management & Plots



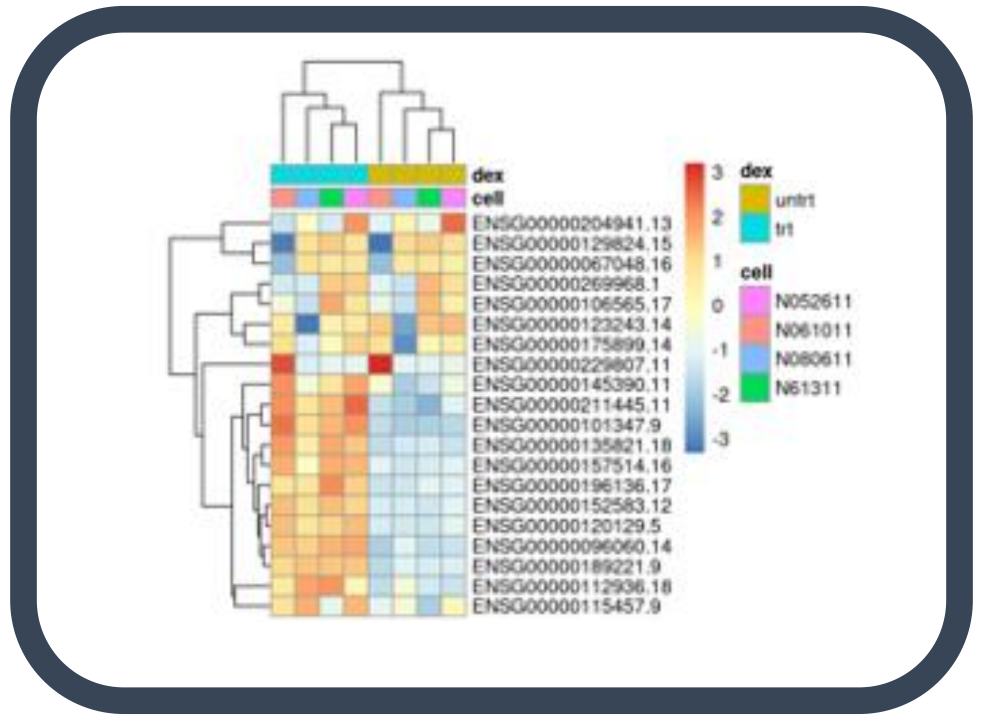
EasyStrata: https://www.uni-regensburg.de/medizin/epidemiologie-praeventivmedizin/genetische-epidemiologie/software/

More Plotting…



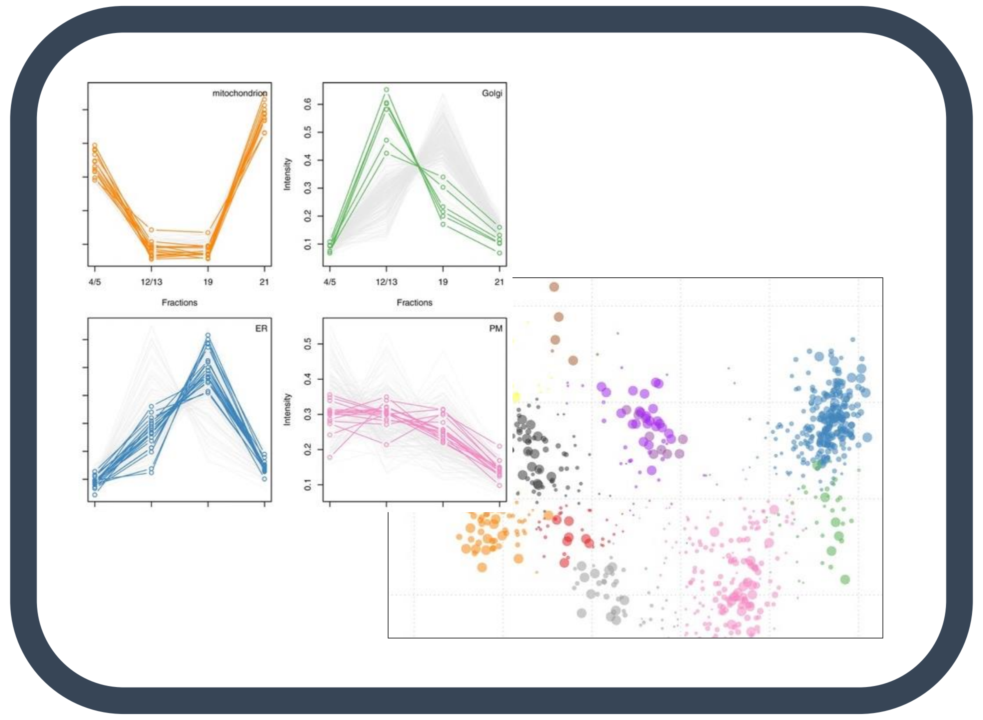
Manhattan and QQ plots: https://cran.r-project.org/web/packages/qqman/vignettes/qqman.htm l

Gene Expression Analysis



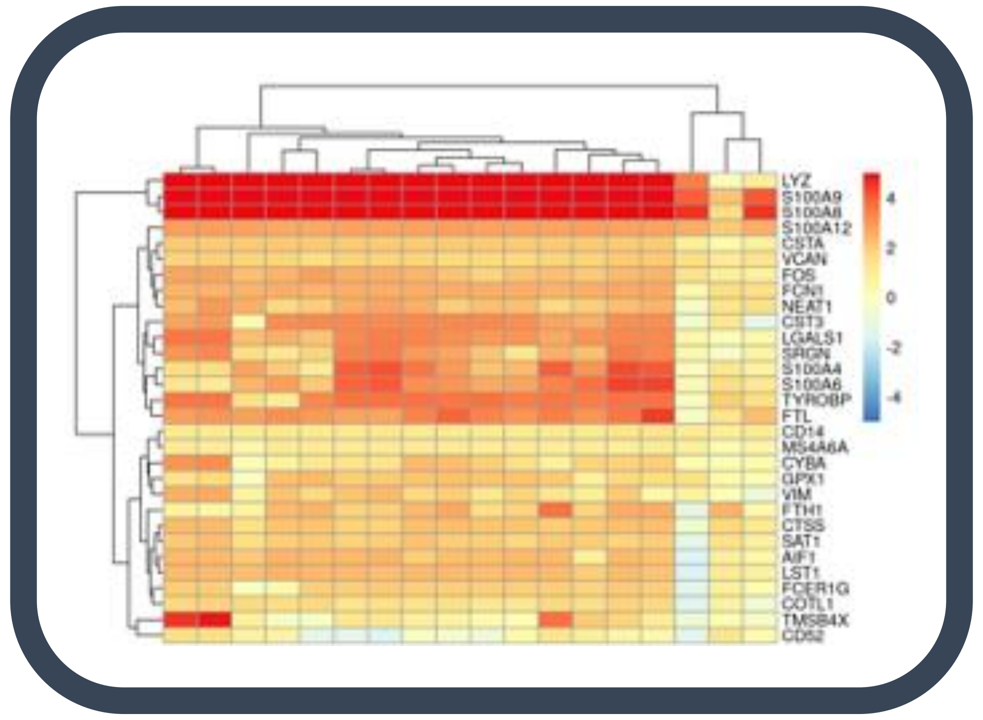
DESeq2, limma, EdgeR, etc.: http://www.bioconductor.org/packages/release/BiocViews.html#___RNASeq

Proteomics Analysis



RforProteomics: https://www.bioconductor.org/packages/release/BiocViews.html#___Proteomics

Single-Cell RNASeq



https://cran.r-project.org/web/packages/e1071/vignettes/svmdoc.pdf

TEASER

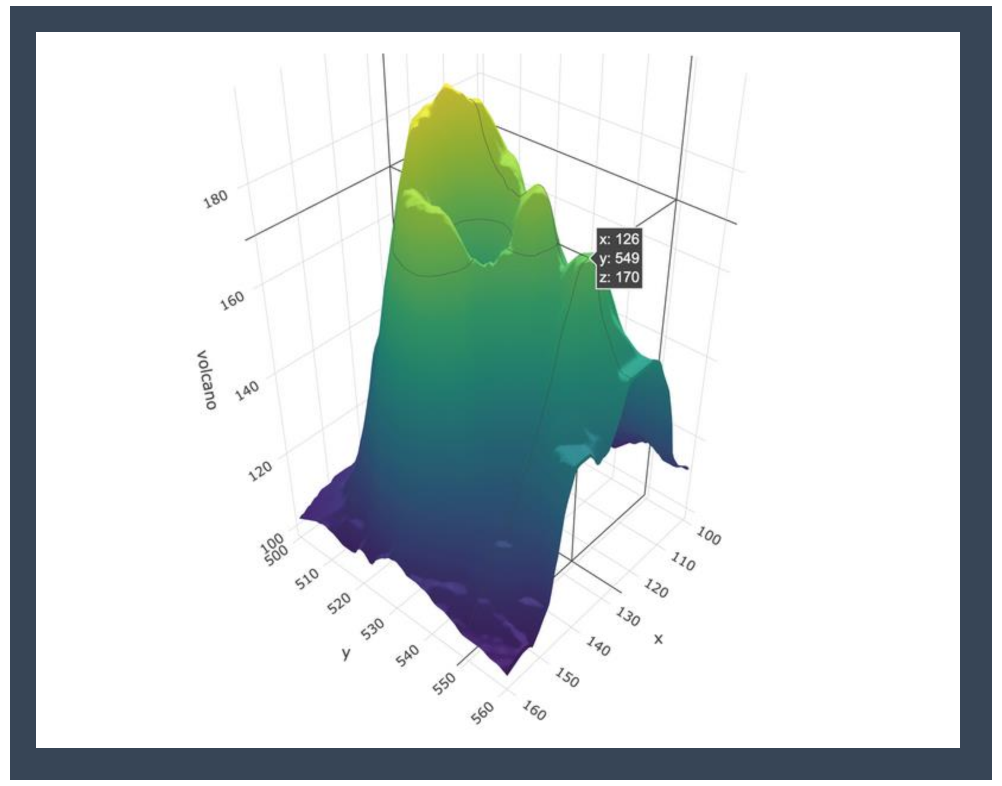
Omics Data

http://www.bioconductor.org/packages/release/BiocViews.html

FROM EXCEL TO R

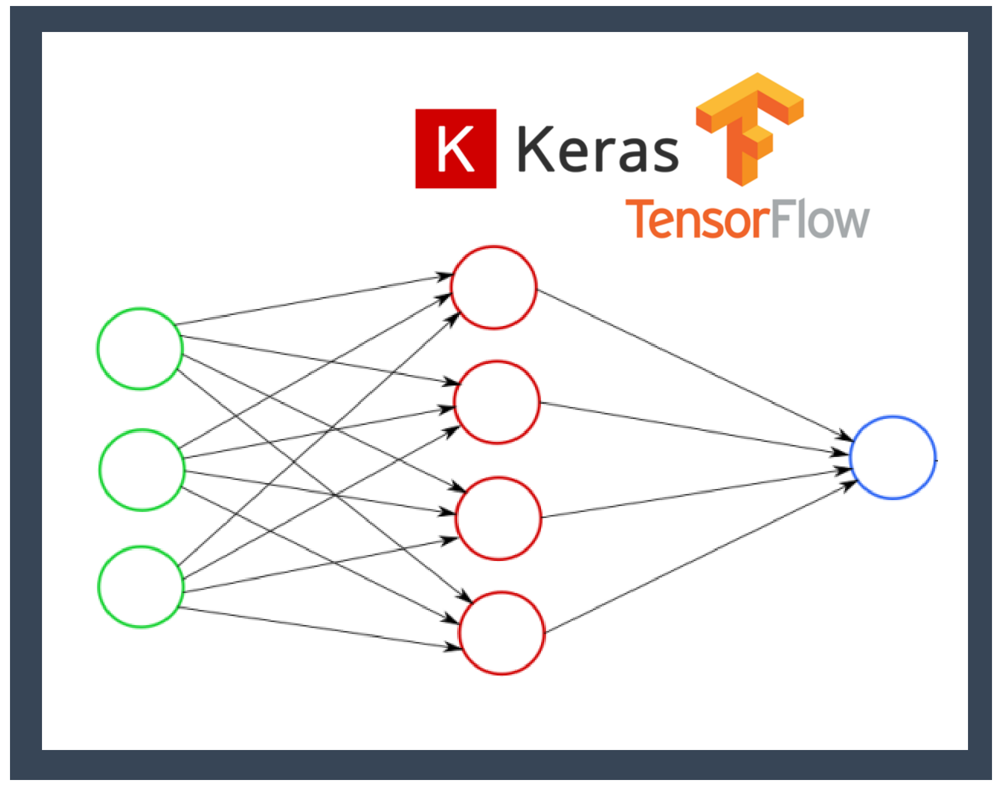
COOL STUFF IN R

PLOTTING IN 3D



https://plotly-r.com/d-charts.html

DEEP LEARNING



https://keras.rstudio.com/
https://tensorflow.rstudio.com/

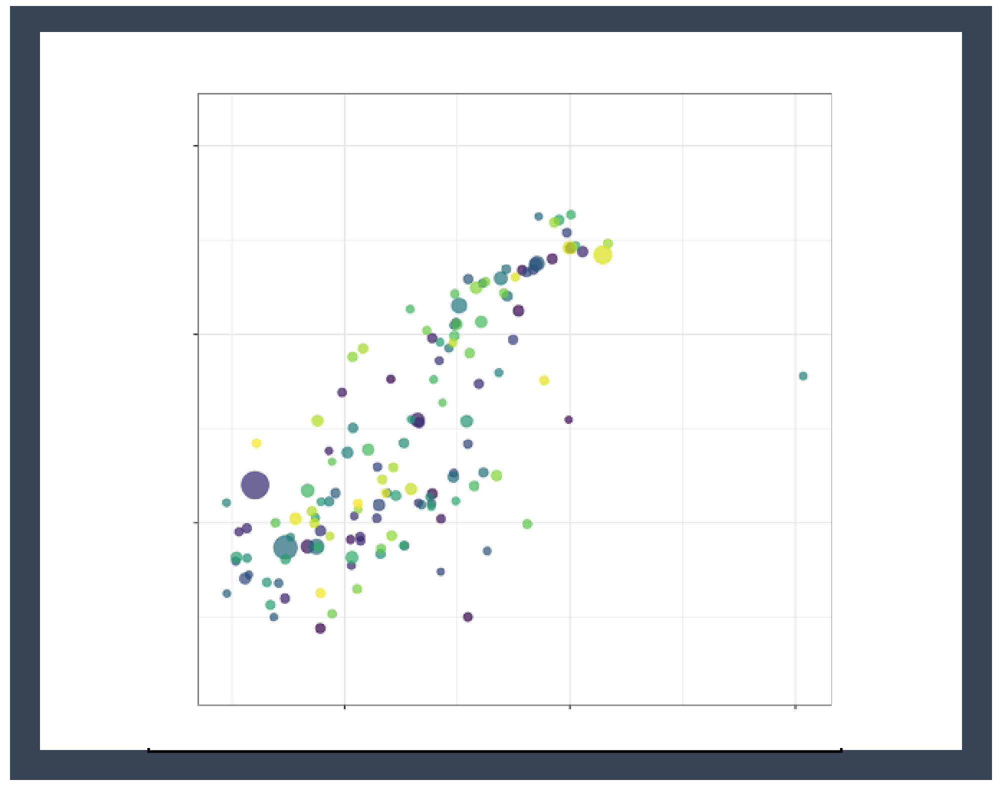
BAYESIAN STATISTICS

https://mc-stan.org/users/interfaces/rstan

WEBPAGE WITH R SHINY

https://shiny.rstudio.com/

INTERACTIVE PLOTS



https://gganimate.com/articles/gganimate.html

MAIL AND MESSAGES

https://github.com/briandconnelly/pushoverr

DATA SCIENCE WITH HEADS DATA LAB

- Drop-in Thursday (13:00-15:00)
- Consultations
- Commission Research & Supervision

- Events and Conferences
- Course Reminders

Read more on HeaDS Website!

https://heads.ku.dk

Flow HeaDS on LinkedIn!

https://www.linkedin.com/company/ucph-heads/

Health Data Science Seminar & Social Networking

TALKING HeaDS

Understanding the Effects of Missense Variants

Using Analyses of Protein Stability and Conservation

with **Kresten Lindorff-Larsen**

27 May at 15:00
Panum, Faculty Club, 16.6.16

REGISTER

https://heads.ku.dk/course/talking-heads-seminar/

Supported by:

THANK YOU FOR LISTENING